

Notes on Fudan QFT lectures

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April 26, 2026

Abstract

These notes are based on the QFT lectures given by Prof. Zhou Yang at Fudan University in 2024 Fall. Almost all the mathematical steps and equations are included, though they look ugly. References may be included implicitly. I use ChatGPT to improve the text. In order to enrich the discussion and mitigate the technical complexities of the physics, I have supplemented the text with data and anecdotes of prominent figures in the field. Typos are inevitable, so if you find any, please let me know.

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Notations and conventions

Units. We will work in natural units, where

$$\hbar = c = 1.$$

Thus in this system,

$$[\text{length}] = [\text{time}] = [\text{energy}]^{-1} = [\text{mass}]^{-1}.$$

The mass (m) of a particle is therefore equal to its rest energy (mc^2), and also its inverse Compton wavelength ($mc/\hbar$).

We use the metric tensor $g = \text{diag}(+1, -1, -1, -1)$, and Einstein summation convention : repeated indices are summed in all cases. The derivative operator are defined as

$$\partial_\mu = \frac{\partial}{\partial x^\mu} = \left(\frac{\partial}{\partial x^0}, \nabla \right),$$

and the totally anti-symmetric tensor $\epsilon^{0123} = +1$. Be careful, in this case $\epsilon_{0123} = -1$.

Fourier Transformations and Distributions. We will often make use of the Heaviside step function $\theta(x)$ and the Dirac delta function $\delta(x)$, defined as follows:

$$\theta(x) = \begin{cases} 0 & x < 0 \\ 1 & x > 0 \end{cases}, \delta(x) = \frac{d}{dx}\theta(x).$$

The delta function $\delta^{(n)}(\mathbf{x})$ in n dimensions, to be concise will be denoted as $\delta^n(\mathbf{x})$, is zero everywhere except at $x = 0$ and satisfies

$$\int d^n x \delta^n(\mathbf{x}) = 1.$$

In Fourier transforms the factors of 2π will always appear with the momentum integral. For example, in four dimensions,

$$f(x) = \int \frac{d^4 k}{(2\pi)^4} \tilde{f}(k) e^{-ikx} dk;$$
$$\tilde{f}(k) = \int \frac{d^4 x}{(2\pi)^4} f(x) e^{ikx} dx.$$

Meanwhile, remember that

$$\int d^4 x f(x) = (2\pi)^4 \delta^4(x).$$

We make use of this quite often.

Quantum Mechanics. We will often work with the Schrödinger wavefunctions of single quantum-mechanical particles. We represent the energy and momentum operators acting on such wavefunctions following the usual conventions:

$$E = i \frac{\partial}{\partial x^0}, \quad \vec{p} = -i \nabla.$$

Discussions of spin in quantum mechanics make use of Pauli matrices:

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Pauli matrices are traceless and anti-symmetric, satisfying the identity

$$\sigma_i \sigma_j = \delta_{ij} + i\epsilon_{ijk} \sigma_k.$$

Electrodynamics. The Coulomb potential of a point charge Q is

$$\Phi = \frac{Q}{4\pi r},$$

and the fine structure constant is

$$\alpha = \frac{e^2}{4\pi} = \frac{e^2}{4\pi\hbar c} \approx \frac{1}{137},$$

the symbol e stands for the charge of the electron, a negative quantity (although the sign rarely matters). We generally work with the relativistic form of Maxwell's equations:

$$\epsilon^{\mu\nu\rho\sigma} \partial_\nu F_{\rho\sigma} = 0, \quad \partial_\mu F^{\mu\nu} = ej^\nu,$$

where

$$A^\mu = (\Phi, \mathbf{A}), \quad F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu.$$

1 Lecture 1

1.1 Why QFT?

QFT is a result of quantum mechanics and special relativity. In 1926, **Klein-Gordon equation** was proposed to describe the motion of a single particle[1][2]:

$$-\hbar^2 \frac{\partial^2}{\partial t^2} \Psi(x, t) = (-\hbar^2 c^2 \nabla^2 + m^2 c^4) \Psi(x, t). \quad (1.1)$$

However, the energy of the particle derived from this is

$$E = \pm \sqrt{|\mathbf{p}|^2 c^2 + m^2 c^4}, \quad (1.2)$$

which can evidently be negative. This terrible fact means that no stable ground state exists. This is **negative-energy problem**. On the other hand, the probability density of the particle is

$$\rho = \frac{i\hbar}{2mc^2} \left(\Psi^* \frac{\partial \Psi}{\partial t} - \frac{\partial \Psi^*}{\partial t} \Psi \right), \quad (1.3)$$

which can also be negative. This is **negative-probability problem**.

The point is that the Klein-Gordon equation involves second derivative to time t . To equally handle time and space in quantum mechanics, one way is to define an operator $\hat{\Phi}(x)$ at each space point x . The set composed of these operators is called **quantum field**. In **Heisenberg picture**, the quantum field is time-dependent:

$$\hat{\Phi}(x, t) = e^{i\hat{H}t/\hbar} \hat{\Phi}(x) e^{-i\hat{H}t/\hbar}. \quad (1.4)$$

Thus, the time- and space-dependent field constitutes a dynamical system. This framework is known as **quantum field theory**, within which the problems we propose are addressed. In this theory, the vacuum represents the ground state of the quantum field, while particles correspond to its excited states. Quantum field theory treats particles and antiparticles symmetrically. The issue of negative energy is resolved, as the Hamiltonian, expressed in terms of creation and annihilation operators, is now positive-definite. Furthermore, ρ is defined as the difference in particle numbers, eliminating the problem of negative probability.

1.2 Lorentz group

As stated, to describe a system moving at high speed, we must incorporate special relativity, where the coordinates of two inertial frames are related through **Lorentz transformation**. Therefore, we shall discuss the Lorentz group, whose elements consist of all possible Lorentz transformations.

We often use the coordinate (x, y) to represent the **rotation** on 2-dimensional space

$$\begin{aligned} x' &= x \cos \theta + y \sin \theta \\ y' &= -x \sin \theta + y \cos \theta, \end{aligned} \quad (1.5)$$

this can be written in a more compact way:

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix} \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix}. \quad (1.6)$$

Using the index notation, it will be even simpler:

$$x_i = R_{ij} x_j. \quad (1.7)$$

For the matrix R , we have

$$RR^T = \mathbb{1}, \quad \det R = 1. \quad (1.8)$$

Indeed, this is exactly the **representation** of special ($\det R = 1$) orthogonal ($RR^T = \mathbb{1}$) group $SO(2)$. We can easily work out the representation for $SO(N)$ group. Under such transformations, length is preserved, i.e. $x^i x_i = x^j x_j$.

It is necessary to generalize this to Minkowski spacetime. Here, we define the 4-vector as:

$$x^\mu = (t, x, y, z) = (x^0, x^1, x^2, x^3). \quad (1.9)$$

The "length" in four-dimensional spacetime will still be preserved:

$$x^\mu g_{\mu\nu} x_\nu = t^2 - x^2 - y^2 - z^2, \quad (1.10)$$

with a non-trivial metric $g_{\mu\nu}$. With foundations, for a Lorentz transformation $x \rightarrow \Lambda x$, we derive

$$x^T g x \equiv |x|^2 \rightarrow x^T \Lambda^T g \Lambda x = x^T g x. \quad (1.11)$$

We arrived at

$$\Lambda^T g \Lambda = g, \quad (1.12)$$

this is **metric-preservation condition**, which Lorentz transformation must satisfy.

We should also pay attention to **boosts** that essentially means taking another reference frame into account. Now we put matrices representing rotations and boosts altogether, there are three matrices for rotations and three for boosts:

$$R_x = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & \cos \theta_x & \sin \theta_x \\ 0 & 0 & -\sin \theta_x & \cos \theta_x \end{pmatrix}, R_y = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos \theta_y & 0 & -\sin \theta_y \\ 0 & 0 & 1 & 0 \\ 0 & \sin \theta_y & 0 & \cos \theta_y \end{pmatrix}, R_z = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos \theta_z & \sin \theta_z & 0 \\ 0 & -\sin \theta_z & \cos \theta_z & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix},$$

$$K_x = \begin{pmatrix} \cosh \beta_x & \sinh \beta_x & 0 & 0 \\ \sinh \beta_x & \cosh \beta_x & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, K_y = \begin{pmatrix} \cosh \beta_y & 0 & \sinh \beta_y & 0 \\ 0 & 1 & 0 & 0 \\ \sinh \beta_y & 0 & \cosh \beta_y & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, K_z = \begin{pmatrix} \cosh \beta_z & 0 & 0 & \sinh \beta_z \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ \sinh \beta_z & 0 & 0 & \cosh \beta_z \end{pmatrix}, \quad (1.13)$$

where $\cosh \beta = 1/\sqrt{1-v^2}$. These transformations compose **Lorentz group** that we denote as $SO(1,3)$.

The comma implies the signature of the metric.

We introduce some discrete transformations called **parity** and **time-reversal**:

$$P : (t, x, y, z) \rightarrow (t, -x, -y, -z)$$

$$T : (t, x, y, z) \rightarrow (-t, x, y, z).$$

Any Lorentz transformation can be classified as either proper orthochronous, or as the product of a proper orthochronous transformation and a discrete transformation, which contains P , T and PT . They compose the group $O(1,3)$. The group $SO(1,3)$ describes **proper orthochronous** transformations with $\det \Lambda = +1$ and $\Lambda_{00} \geq +1$, it is the subgroup of $O(1,3)$.

Quantities are classified by how they transform along with coordinate transformations. Scalars are invariant under Λ transformations while vectors received an extra factor: $x_\mu \rightarrow \Lambda^\nu_\mu x^\nu$. The transformation for tensor is more complicated: $T_{\mu\nu} \rightarrow \Lambda^\alpha_\mu \Lambda^\beta_\nu T_{\alpha\beta}$.

The inner product of two vectors in four-dimensional spacetime is

$$V^\mu g_\mu^\nu W_\nu \equiv V_0 W_0 - V_1 W_1 - V_2 W_2 - V_3 W_3. \quad (1.14)$$

This should be invariant under Lorentz transformations. According to the sign of the inner product of two vectors, we classify the spacetime interval as

$$\begin{cases} V^\mu V_\mu > 0 & \text{timelike} \\ V^\mu V_\mu = 0 & \text{lightlike} \\ V^\mu V_\mu < 0 & \text{spacelike} \end{cases}.$$

1.3 Harmonic oscillator

The harmonic oscillator is described by the second-order ordinary derivative equation of x :

$$m \frac{d^2 x}{dt^2} + kx = 0, \quad (1.15)$$

with k being a constant. Defining angular frequency as $\omega = \sqrt{k/m}$, the general solution is

$$x(t) = A \sin \omega t + B \cos \omega t. \quad (1.16)$$

Using knowledge of complex analysis, the solution can also be expressed as

$$x(t) = \frac{1}{\sqrt{2m\omega}} (a e^{-i\omega t} + a^* e^{i\omega t}), \quad (1.17)$$

where a^* is the Hermitian conjugate of a . This will be our foundation for later discussions on field quantization.

2 Lecture 2

2.1 Classical field theory

The dynamics of the field is governed by Lagrangian, which is a function of $\phi(x, t)$, $\dot{\phi}(x, t)$, and $\nabla\phi(x, t)$. In local field theory, Lagrangian is written as

$$L(t) = \int d^3x \mathcal{L}(\phi_a, \partial_\mu \phi_a). \quad (2.1)$$

In the context, we simply refer to Lagrangian density as Lagrangian, and it is Lorentz invariant. "Local" means that the Lagrangian depends on just one spacetime point. The action is the integral of Lagrangian in configuration space

$$S = \int dt \int d^3x \mathcal{L}. \quad (2.2)$$

Note that L depends on q and $\dot{q}$ but not on $\ddot{q}$.

By varying $\phi \rightarrow \phi + \delta\phi$, the variation of action is

$$\begin{aligned} \delta S &= \int d^4x \left[\frac{\partial \mathcal{L}}{\partial \phi_a} \delta\phi_a + \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi_a)} \delta(\partial_\mu \phi_a) \right] \\ &= \int d^4x \left[\frac{\partial \mathcal{L}}{\partial \phi_a} - \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi_a)} \right) \right] \delta\phi_a + \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi_a)} \right) \delta\phi_a. \end{aligned} \quad (2.3)$$

The last term can be turned into a surface integral over the boundary of the four-dimensional space-time region of integration. If we restrict our consideration to deformations $\delta\phi$ that vanish on the spatial boundary of the region as well, then the surface term is zero. Since the integral must vanish for arbitrary $\delta\phi$, we arrive at the

$$\frac{\partial \mathcal{L}}{\partial \phi_a} - \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi_a)} \right) = 0, \quad (2.4)$$

which gives the classical equation of motion (EOM) for field ϕ . If the Lagrangian contains more than one field, there is one such equation for each. We define $p \equiv \partial \mathcal{L} / \partial \dot{q}$, and Hamiltonian is $H = \sum p \dot{q} - L$. Conjugate momentum density is written as $\Pi \equiv \partial \mathcal{L} / \partial \dot{\phi}(x, t)$. Passing to the continuum, we have

$$H = \int d^3x [\pi(x) \dot{\phi}(x) - \mathcal{L}] \equiv \int d^3x \mathcal{H}. \quad (2.5)$$

As a simple example, consider the theory of free scalar field:

$$\mathcal{L} = \frac{1}{2}(\partial_\mu \phi)^2 - \frac{1}{2}(\nabla\phi)^2 - \frac{1}{2}m^2\phi^2.$$

Its equation of motion reads

$$(\square + m^2)\phi = 0,$$

where the $\square \equiv \partial_\mu \partial^\mu$ is known as the d'Alembertian.

2.2 Locality

Locality is of importance in classical field theory, it essentially means that EOMs contain only field & derivatives at the *same* spacetime point. A quantity should be completely determined by its neighbor-

hood. You should think about examples like free scalar field, Maxwell equations, general relativity or Dirac's equation. Something violates locality like $\mathcal{L} = \int d^3x d^3y \phi(x, t)\phi(y, t)$ is not allowed. The Lagrangian must be Lorentz-invariant so that the theory respects special relativity, which implies that terms should be formed from scalar combinations of fields and derivatives. Locality is the reason to introduce fields.

2.3 Symmetry

Consider an infinitesimal continuous transformation on the field ϕ

$$\phi(x) \rightarrow \phi'(x) = \phi(x) + \partial\Delta\phi(x), \quad (2.6)$$

if this leaves the action invariant, the field has a symmetry under such transformations. We postpone more detailed discussions as the concept of symmetry can be found discussed everywhere, even beyond the quantum field theory.

3 Lecture 3

3.1 Free fields quantization

There are five steps necessary to quantize the free field:

- (1) Write down equation of motion and equation that operators satisfy;
- (2) Find the solution to the equation;
- (3) Promote integration constants to operators;
- (4) Impose quantization conditions;
- (5) Build up Hilbert space.

As listed above, we take free field as an example. The equation that the field satisfies is

$$\square\phi = (\partial^2 - \nabla^2)\phi = 0. \quad (3.1)$$

Recall the case of harmonic oscillator, where the equation $\ddot{x} + \omega^2 x = 0$ is what we are familiar with, which reminds us that we should try a solution like $a_p(t)e^{ip \cdot x}$. Plugging in this, we arrived at

$$(\partial_t^2 + \vec{p} \cdot \vec{p})a_p(t) = 0. \quad (3.2)$$

Next, we work out general solution

$$\phi(x, t) = \int \frac{d^3p}{(2\pi)^3} (a_p(t)e^{i\vec{p} \cdot \vec{x}} + a_p^\dagger(t)e^{-i\vec{p} \cdot \vec{x}}). \quad (3.3)$$

This is a plane-wave expansion in form of Fourier integral, it expands the field $\phi(x, t)$ to sum of infinitely many momentum modes. The time-dependence in a_p and $a_p^\dagger$ operator can be realized by four-vector in the exponent:

$$\phi(x, t) = \int \frac{d^3p}{(2\pi)^3} (a_p e^{ip \cdot x} + a_p^\dagger e^{-ip \cdot x}). \quad (3.4)$$

As in special relativity, $p^\mu \equiv (\omega, \vec{p})$.

We have done the basic processes of field quantization, each $\vec{p}$ is considered as a harmonic oscillator, a_p and $a_p^\dagger$ are corresponding annihilation and creation operators, satisfying **canonical quantization conditions**

$$[a_k, a_p^\dagger] = (2\pi)^3 \delta^3(\vec{p} - \vec{k}), \quad [a_k, a_p] = 0, \quad [a_k^\dagger, a_p^\dagger] = 0. \quad (3.5)$$

In Chapter 10, we will introduce a more general approach to field quantization: the path integral, a functional method.

3.2 Fock space

To build up the Hilbert space, we define the vacuum state $|0\rangle$ that would be annihilated by a_p , i.e. $a_p |0\rangle = 0$. We act $a_p^\dagger$ on $|0\rangle$, the Hilbert space can be constructed:

$$\mathcal{H} = \oplus_{n \geq 0} \mathcal{H}_n, \quad \mathcal{H}_n = \text{span}\{a_p^\dagger, \dots, (a_p^\dagger)^n |0\rangle\}. \quad (3.6)$$

We call this **Fock space**. Hereafter, all the five steps are completed.

The canonical quantization conditions are actually equivalent to

$$[\hat{\phi}(x, t), \hat{\pi}(x, t)] = i\delta^3(x - x'), \quad [\hat{\phi}(x, t), \hat{\phi}(x', t)] = 0, \quad [\hat{\pi}(x, t), \hat{\pi}(x', t)] = 0. \quad (3.7)$$

These are equal-time commutation relation that will be quite often used in the following chapter, make sure that they are familiar to you. The process of quantizing an arbitrary number of particles instead of a single particle is often also called **second quantization**.

Remark 1. • $\omega = \sqrt{|p|^2 + m^2}$, $(\square + m^2)\phi = 0$

• The Hamiltonian can now be expressed in terms of a_p and $a_p^\dagger$

$$H = \int \frac{d^3p}{(2\pi)^3} \omega_p \left(a_p^\dagger a_p + \frac{1}{2} [a_p, a_p^\dagger] \right) \quad (3.8)$$

With all these preparations, we have more commutation relation:

$$\begin{aligned} [H, \phi(x, t)] &= \int \frac{d^3p}{(2\pi)^3} \int \frac{d^3k}{(2\pi)^3} \frac{1}{\sqrt{2\omega_k}} \left[\omega_p \left(a_k + \frac{V}{2} \right), a_k e^{-ik \cdot x} + a_k^\dagger e^{ik \cdot x} \right] \\ &= -i\partial_t \phi(x, t). \end{aligned} \quad (3.9)$$

Momentum is now

$$\vec{P} = - \int d^3x \pi \nabla \phi = \int \frac{d^3x}{(2\pi)^3} \vec{p} a_p^\dagger a_p, \quad (3.10)$$

and

$$\langle 0 | \vec{p} | 0 \rangle = 0, \quad \sqrt{2\omega_p} a_p^\dagger | 0 \rangle \equiv | \vec{p} \rangle, \quad (3.11)$$

where the $\sqrt{2\omega_p}$ is a normalization factor.

4 Lecture 4

4.1 Propagators

Let's study commutator $[\phi(x), \phi(y)]$ a little further. Since it is a c-number, we can write $[\phi(x), \phi(y)] = \langle 0 | [\phi(x), \phi(y)] | 0 \rangle$. Assuming that $x^0 > y^0$, this can be written as a four-dimensional integral:

$$\begin{aligned} [\phi(x), \phi(y)] &= \int \frac{d^3p}{(2\pi)^3} \frac{1}{2\omega_p} (e^{-ip \cdot (x-y)} - e^{ip \cdot (x-y)}) \\ &= \int \frac{d^3p}{(2\pi)^3} \left(\frac{1}{2\omega_p} e^{-ip \cdot (x-y)} \Big|_{p_0=\omega_p} + \frac{1}{-2\omega_p} e^{-ip \cdot (x-y)} \Big|_{p_0=-\omega_p} \right) \\ &= \int \frac{d^3p}{(2\pi)^3} \int dp_0 \frac{-e^{-ip \cdot (x-y)}}{p^2 - m^2}. \end{aligned} \quad (4.1)$$

Constructing the following contour, one can evaluate the integral above. For the case of $x^0 < y^0$, we close the contour from above and do the same thing. Eventually, we obtain the expression

$$D_R(x-y) = \theta(x^0 - y^0) \langle 0 | [\phi(x), \phi(y)] | 0 \rangle \equiv \int \frac{d^4p}{(2\pi)^4} \frac{-e^{-ip \cdot (x-y)}}{p^2 - m^2}. \quad (4.2)$$

The definition above relies on the contour we choose. We can act $\partial^2 + m^2$ on the both sides (by yourself!) and verify that this quantity is the Green's function of Klein-Gordon propagator.

Clearly, the choice of contour is not unique; we may select an alternative one. In this way, one can write down the correlator as

$$D_F(x-y) \equiv \frac{d^4p}{(2\pi)^4} \frac{i}{p^2 - m^2 + i\epsilon} e^{-ip \cdot (x-y)}. \quad (4.3)$$

The propagator above is **Feynman propagator**, explicitly marked with a subscript. Note that we have added the infinitesimal imaginary part $i\epsilon$ in the denominator, by this we integrate along a contour that is rotated slightly away from the real axis. Therefore we no longer need to detour the poles on the real axis. The $i\epsilon$ prescription is equivalent to this.

$$D_F(x-y) = \theta(x^0 - y^0) \langle 0 | \phi(x) \phi(y) | 0 \rangle + \theta(y^0 - x^0) \langle 0 | \phi(y) \phi(x) | 0 \rangle = \langle 0 | T \phi(x) \phi(y) | 0 \rangle, \quad (4.4)$$

where T is called **time-ordering operator**, which moves the field with later time to the left.

We should carefully consider the interpretation of **causality**, which states that a measurement at one point may influence a measurement at another point. This is the reason why $[\phi(x), \phi(y)]$ gives a nonzero value. In particular,

$$\langle 0 | [\phi(x), \phi(y)] | 0 \rangle = D(x-y) - D(y-x). \quad (4.5)$$

For spacelike interval $(x^0 - y^0)^2 < 0$, the matrix element above is zero. Otherwise (for timelike or lightlike interval), it would not be zero. It's a detection of causality in some sense.

Let us examine how the correlation function is related to the Green's function. We choose retarded Green's function and act operator $\square + m^2$ on it:

$$\begin{aligned} (\square + m^2) D_R(x-y) &= \partial^2 \theta(x^0 - y^0) \langle 0 | [\phi(x), \phi(y)] | 0 \rangle + 2\partial_\mu \theta(x^0 - y^0) \partial^\mu \langle 0 | [\phi(x), \phi(y)] | 0 \rangle \\ &\quad + (\partial^2 + m^2) \theta(x^0 - y^0) \langle 0 | [\phi(x), \phi(y)] | 0 \rangle \\ &= -\delta(x^0 - y^0) \langle 0 | [\pi(x), \phi(y)] | 0 \rangle + 2\delta(x^0 - y^0) \langle 0 | [\pi(x), \phi(y)] | 0 \rangle \\ &= -i\delta^4(x-y). \end{aligned} \quad (4.6)$$

The term on the second line vanishes due to the restriction of delta function. If you are confused, check the **Notations and Conventions** again. Get to the point, we proved that D_R is the Green's function to the operator $\square + m^2$.

4.2 Interacting theory

As researchers in an experimental discipline, physicists are responsible for validating their theories through experimentation. It is essential to focus on interactions, particularly in high-energy physics, where this is typically achieved using colliders. In these setups, thousands of particle scatterings occur. The most commonly measured quantities in such experiments are the **cross section** and **decay rates**. We will establish their connection to a more fundamental quantity: the S-matrix.

We prepare initial states for incident particles $|i\rangle = |p_1, p_2\rangle$, and final states $|f\rangle = |p_3, \dots, p_n\rangle$. At a very long time before and after they interact, they are viewed as free. They interact with each other only at the collisions. Then the amplitude is given by matrix element

$$\langle f|U(+\infty, -\infty)|i\rangle_H = \langle f, +\infty|i, -\infty\rangle_S, \quad (4.7)$$

where we use different subscripts to mark Heisenberg or Schrödinger picture. Heisenberg picture is easier for us to construct our theory.

4.3 S-matrix

Actually, we define **S-matrix** exactly in this way:

$$S_{fi} = \langle f|U(+\infty, -\infty)|i\rangle_H. \quad (4.8)$$

S-matrix is unitary: $SS^\dagger = 1$. As you see above, we perceive the initial and final states as column and row indices, and thus we constructed a matrix. If particles do not interact at all, S is simply the identity operator. To isolate the interesting part of the S-matrix—that is, the part due to interactions—we define the T-matrix by

$$S = \mathbb{1} + iT. \quad (4.9)$$

The matrix element of S should reflect 4-momentum conservation. Thus S or T should always contain a factor $\delta^4(p_1 + p_2 - \sum p_f)$. We define the **invariant matrix element** $\mathcal{M}$, by

$$\langle p_1, p_2|iT|p_3, \dots, p_f\rangle = (2\pi)^4 \delta^4(p_1 + p_2 - \sum p_f) \cdot i\mathcal{M}(p_1, p_2 \rightarrow p_f). \quad (4.10)$$

It is also called **scattering amplitude**.

S-matrix is directly related to correlation functions, this is the reason why we particularly care about correlation functions, so far.

Without proof, we claim the **Lehmann-Symanzik-Zimmermann (LSZ) reduction formula**:

$$\langle f|S|i\rangle = \left(\int d^4x_1 e^{-ip_1 \cdot x_1} (\square_1^2 + m^2) \right) \dots \left(\int d^4x_n e^{-ip_n \cdot x_n} (\square_n^2 + m^2) \right) \times \langle \Omega|T\phi(x_1) \dots \phi(x_n)|\Omega\rangle. \quad (4.11)$$

It's a sophisticated but distinct formula. In more physical terms, the LSZ reduction formula tells you how to extract the real, observable scattering events from the more abstract theoretical framework involving interacting fields and Green's functions. We will make use of it in the next Chapter.

In any case, I don't know of anyone who has ever needed to use the details of LSZ for anything. So I encourage you to skim the proof on your own, but don't stress out about it if you can't master

it on first pass. You can study it again in the future if you want to—it's not going anywhere.
—Matthew Schwartz

5 Lecture 5

5.1 Schwinger-Dyson equation

We have derived the time-ordered correlation functions for interacting theory, but how to compute? It is meaningful to show

$$(\square + m^2) \langle \Omega | T \phi(x) \phi(x') | \Omega \rangle = \langle \Omega | T (\square + m^2) \phi(x) \phi(x') | \Omega \rangle - i\hbar \delta^4(x - x'), \quad (5.1)$$

the second term on the R.H.S is called **contact term**.

Proof. According to the definition of time-ordering operator,

$$\begin{aligned} \partial_t \langle \Omega | T \phi(x) \phi(x') | \Omega \rangle &= \partial_t (\langle \Omega | \phi(x) \phi(x') | \Omega \rangle \theta(t - t') + \langle \Omega | \phi(x') \phi(x) | \Omega \rangle \theta(t' - t)) \\ &= \langle \Omega | T (\partial_t \phi(x) \phi(x')) | \Omega \rangle + \delta(t - t') \langle \Omega | [\phi(x), \phi(x')] | \Omega \rangle, \end{aligned} \quad (5.2)$$

where the last term vanishes due to the restriction of delta function. Act it once more:

$$\begin{aligned} \partial_t^2 \langle \Omega | T \phi(x) \phi(x') | \Omega \rangle &= \langle \Omega | T (\partial_t^2 \phi(x) \phi(x')) | \Omega \rangle + \delta(t - t') \langle \Omega | [\partial_t \phi(x), \phi(x')] | \Omega \rangle \\ &= \langle \Omega | T (\partial_t^2 \phi(x) \phi(x')) | \Omega \rangle - i\delta^4(x - x'). \end{aligned} \quad (5.3)$$

Note that we have used equal-time commutation relation (3.7) in the second step, causing the last term to no longer vanish. Since the time derivative operator ∂_t and constant can be moved into or out of the matrix element freely, the equation (5.1) is proved. Generalize this into the case of n fields:

$$\square \langle \phi(x) \phi(x_1) \cdots \phi(x_n) \rangle = \langle \square \phi(x) \phi(x_1) \cdots \phi(x_n) \rangle - i\hbar \sum_j \delta^4(x - x_j) \langle \phi(x) \cdots \phi(x_{j-1}) \phi(x_{j+1}) \cdots \phi(x_n) \rangle. \quad (5.4)$$

Generally, the Lagrangian for scalar field can be set as

$$\mathcal{L} = -\frac{1}{2} \phi (\square + m^2) \phi + \mathcal{L}_{\text{int}}[\phi]. \quad (5.5)$$

The equation of motion reads

$$(\square + m^2) \phi = \mathcal{L}'[\phi], \quad (5.6)$$

where

$$\mathcal{L}'_{\text{int}}[\phi] \equiv \frac{d}{d\phi} \mathcal{L}_{\text{int}}.$$

Substitute this into equation (5.4), we have

$$\square \langle \phi(x) \phi(x_1) \cdots \phi(x_n) \rangle = \langle \mathcal{L}'_{\text{int}} \phi(x) \phi(x_1) \cdots \phi(x_n) \rangle - i\hbar \sum_j \delta^4(x - x_j) \langle \phi(x) \cdots \phi(x_{j-1}) \phi(x_{j+1}) \cdots \phi(x_n) \rangle. \quad (5.7)$$

In this equation, the angle-brackets denote a time-ordered correlation function in which derivatives on $\phi(x)$ are placed outside the time-ordering symbol. These are **Schwinger-Dyson equations** for $n + 1$ fields, which reveals the distinction between quantum and classical system. For classical fields, the equal-time commutation relation no longer holds such that contact terms vanish.

Now, let's derive correlation functions from Schwinger-Dyson equations. First, to simplify notations, we claim abbreviations: $\phi_x \equiv \phi(x)$, $\phi_j \equiv \phi(x_j)$, $\delta_{xi} \equiv \delta(x - x_i)$ and $D_{ij} = D_{ji} \equiv D(x_i - x_j)$. In this way the Green's function is written in a simple form

$$\square_i D_{xi} = -i\delta_{xi}. \quad (5.8)$$

The free field two-point function

$$\langle \phi_1 \phi_2 \rangle = \int d^4x \delta_{x1} \langle \phi_x \phi_2 \rangle = i \int d^4x \square_x D_{x1} \langle \phi_x \phi_2 \rangle = i \int d^4x D_{x1} \square_x \langle \phi_x \phi_2 \rangle = \int d^4x D_{x1} \delta_{x2} = D_{12}. \quad (5.9)$$

We applied the definition of δ -function in step one, and Green's function (5.8) in step two and four. We do the same thing for a four-point function

$$\begin{aligned} \langle \phi_1 \phi_2 \phi_3 \phi_4 \rangle &= i \int d^4x D_{x1} \square_x \langle \phi_1 \phi_2 \phi_3 \phi_4 \rangle \\ &= \int d^4x D_{x1} (\delta_{x2} \langle \phi_3 \phi_4 \rangle + \delta_{x3} \langle \phi_2 \phi_4 \rangle + \delta_{x4} \langle \phi_2 \phi_3 \rangle) \\ &= D_{12} D_{34} + D_{13} D_{24} + D_{14} D_{23}. \end{aligned} \quad (5.10)$$

This result is in agreement with what we obtained from Wick's theorem.

For a interacting theory with $\mathcal{L}_{\text{int}} = \phi^3/6$, following the Schwinger-Dyson equation,

$$\begin{aligned} \langle \phi_1 \phi_2 \rangle &= i \int d^4x D_{1x} \left(\frac{g}{2!} \langle \phi_x^2 \phi_2 \rangle \right) \\ &= D_{12} - \frac{g}{2} \int d^4x \int d^4y D_{x1} D_{y2} \square_y \langle \phi_x^2 \phi_y \rangle \\ &= D_{12} - \frac{g^2}{4} \int d^4x \int d^4y D_{x1} D_{y2} \langle \phi_x^2 \phi_y^2 \rangle + ig \int d^4x D_{1x} D_{2x} \langle \phi_x \rangle. \end{aligned} \quad (5.11)$$

Freeman Dyson: A Life of Curiosity and Unconventional Wisdom

Freeman Dyson (1923-2020), one of the most influential theoretical physicists of the 20th century, was celebrated for his groundbreaking work in quantum electrodynamics (QED), nuclear physics, and mathematics. Born in 1923 in England, Dyson was an intellectually gifted child who pursued a career in science, driven not by fame but by a deep curiosity about the world. His early education at Cambridge led him to work with luminaries like Paul Dirac, and after World War II, he moved to Princeton, where he became part of a vibrant intellectual community that included Einstein and von Neumann.

Dyson's most famous contribution came in his work on quantum electrodynamics, where he developed the renormalization technique, a key method for calculating interactions between light and matter. His ability to collaborate across disciplinary boundaries led him to work with the likes of Richard Feynman, with whom he initially disagreed but later grew to admire. Dyson's ability to challenge traditional thinking and explore new approaches set him apart in the scientific world.

What truly defined Dyson, however, was his broad intellectual curiosity. He was not confined to a single field of study. He explored concepts as diverse as space travel, extraterrestrial life, and environmental concerns, making significant contributions to both theoretical physics and the practical future of humanity. One of his most famous ideas was the concept of a **Dyson Sphere**: a hypothetical structure that could surround a star to capture its energy, a theory that still captures the imagination of those who speculate on the existence of advanced extraterrestrial civilizations.

Dyson was known for his humility and disinterest in recognition. Unlike many of his peers, he was uninterested in the pursuit of fame or accolades. He never sought a Nobel Prize, and even his work on key theories like the Majorana fermion went uncelebrated during his lifetime. Dyson's attitude was to do the work for the joy of discovery, not for the spotlight.

An anecdote that illustrates Dyson's unconventional approach to life was his resistance to the academic establishment. He once said that he preferred to stay in the background, working quietly on projects that intrigued him, often outside the mainstream of scientific thought. His writings were similarly accessible, blending deep theoretical insights with an engaging, conversational style that

made complex ideas understandable to non-specialists.

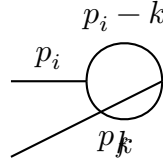
Freeman Dyson's legacy is not just in his theories but in his approach to life. His work on QED, his thoughts on the future of humanity, and his philosophical reflections on science set him apart as a scientist who thought outside the box. Dyson's enduring influence comes not from seeking recognition but from a lifelong commitment to the pursuit of knowledge and the courage to explore ideas that challenged conventional wisdom.

5.2 Feynman rules

Following the last section, we summarize the Feynman rules in configuration space:

- (1) Start with points x_i for which correlation functions is measured. Draw a line from each point;
- (2) A line could either contract to an existing line and give a Feynman propagator, or it split according to the interaction;
- (3) Sum all diagrams with all lines contracted and all internal vertices integrated over;
- (4) Drop out all the $n!$ factors in the coefficient of the interaction, and divide by geometric symmetry factor.

We also care about momentum-space Feynman rules. Let's begin with an example:



The amplitude of this diagram is

$$\mathcal{T}_1 = -\frac{g^2}{2} \int d^4x d^4y D_{1x} D_{xy}^2 D_{y2}. \quad (5.12)$$

Recall that by Fourier transformation, the Feynman propagator is written as:

$$D_{xy} = \int \frac{d^4p}{(2\pi)^4} \frac{i}{p^2 + i\epsilon} e^{ip \cdot (x-y)}. \quad (5.13)$$

Therefore,

$$\begin{aligned} \mathcal{T}_1 &= -\frac{g^2}{2} \int d^4x \int d^4y \int \frac{d^4p_1}{(2\pi)^4} \int \frac{d^4p_2}{(2\pi)^4} \int \frac{d^4p_3}{(2\pi)^4} \int \frac{d^4p_4}{(2\pi)^4} \\ &\quad \times \exp(ip_1 \cdot (x_1 - x)) \exp(ip_2 \cdot (y - x_2)) \exp(ip_3 \cdot (x - y)) \exp(ip_4 \cdot (x - y)) \\ &\quad \times \frac{i}{p_1^2 + i\epsilon} \frac{i}{p_2^2 + i\epsilon} \frac{i}{p_3^2 + i\epsilon} \frac{i}{p_4^2 + i\epsilon} \\ &= -(2\pi)^4 \delta(p_1 - p_2) \cdot \frac{g^2}{2} \int \frac{d^4k}{(2\pi)^4} \int \frac{d^4p_1}{(2\pi)^4} \int \frac{d^4p_2}{(2\pi)^4} \\ &\quad \times \exp(ip_1 \cdot x_1) \exp(ip_2 \cdot x_2) \frac{i}{p_1^2 + i\epsilon} \frac{i}{p_2^2 + i\epsilon} \frac{i}{(p_1 - k)^2 + i\epsilon} \frac{i}{k^2 + i\epsilon}. \end{aligned} \quad (5.14)$$

For simplicity, we omitted mass terms in the denominator of the equations above. The integrals over x and y give us delta functions, thus we killed p_1 and replaced p_4 with k .

Next we take advantage of LSZ reduction formula (4.11), to relate our result with S-matrix element. But we solely need a simpler one:

$$\langle f|S|i\rangle = \left(\int (-i) d^4x_1 \exp(-ip_i \cdot x_1) p_i^2 \int (-i) d^4x_2 \exp(-ip_f \cdot x_2) p_f^2 \right) \times \langle \Omega | T \phi(x_1) \phi(x_2) | \Omega \rangle . \quad (5.15)$$

Obviously $p_i = p_f$. Plug $\mathcal{T}_1$ into time-ordered correlation function in (5.15) and finish the calculation remaining by yourself. According to the result, we obtain the momentum-space Feynman rules.

6 Lecture 6

6.1 Wigner's theorem

In flat spacetime, the Poincaré group is essentially the Lorentz group (comprising rotations and boosts) with translations appended. This is a fundamental symmetry that all physical systems should respect. On the other hand, our universe is populated by particles of various types. A particle can be defined as a set of states that transform among themselves under a Poincaré transformation, $|\psi'\rangle = U|\psi\rangle$, where the unitarity of U is evident. The collection of states $|\psi\rangle$ forms a representation of the Poincaré group. If no subset of the states transforms among themselves, it is referred to as an irreducible representation.

In 1939, Wigner's theorem was proposed: The irreducible unitary representations are classified by quantum numbers m and J , where m is a positive real number and J is a positive half integer, representing mass and angular momentum respectively.

When $J > 0$, for each value of momentum $p^2 = m^2$, there are $2J + 1$ independent states if $m > 0$; Otherwise, there are merely 2 states. When $J = 0$, there is only 1 state for each p^2 , regardless of the value of m . You can check the theorem with the particles of different spins (spin-0 scalar field or spin-1 electromagnetic field, etc.). Textbook [3] is well-known for its discussion on this topic.

Eugene Wigner and Manhattan Project

Eugene Wigner (1902-1995) was a Hungarian-American theoretical physicist known for his significant contributions to quantum mechanics, the theory of symmetries, and nuclear physics. He was awarded the Nobel Prize in Physics in 1963 for his work in theoretical physics.

Although not politically active, Eugene Wigner played a significant role in the development of nuclear weapons during World War II. On August 2, 1939, he participated in the meeting with Leó Szilárd and Albert Einstein that led to the famous Einstein-Szilárd letter. This letter warned President Franklin D. Roosevelt about the possibility of Nazi Germany developing an atomic bomb and prompted the creation of the U.S. Advisory Committee on Uranium to investigate atomic weaponry. Wigner feared that Germany might build the bomb first and even went so far as to refuse to have his fingerprints taken, fearing they could be used to track him down if Germany won the war.

In 1941, Wigner married Mary Annette Wheeler, a professor of physics. They had two children, and Wigner remained married to her until her death in 1977.

During the Manhattan Project, Wigner led a team tasked with designing nuclear reactors to produce weapons-grade plutonium. His team successfully designed a reactor and witnessed the first self-sustaining nuclear chain reaction at Chicago Pile One (CP-1) in December 1942. Wigner, however, was initially disappointed when DuPont was assigned the responsibility for detailed reactor design. Nevertheless, their design adjustments ultimately saved the project when neutron poisoning became an issue.

Wigner did not regret his involvement in building the atomic bomb, expressing that the project could have been completed sooner, which might have had significant geopolitical implications during the war. One of his key contributions to nuclear science was the discovery of the "Wigner effect," a swelling of graphite caused by neutron radiation, which became a problem for post-war reactors but was later overcome.

Additionally, Wigner, together with Leonard Eisenbud, developed the Wigner-Eisenbud R-matrix theory, a significant advancement in nuclear reactions, published in 1947.

6.2 Spin 1 and gauge invariance

Let's talk something about electromagnetic field, it obeys the celebrated Maxwell's equation. The Lagrangian reads

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu}, \quad (6.1)$$

with energy-momentum tensor to be

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu. \quad (6.2)$$

Therefore, we obtained the equation of motion

$$\partial_\mu F^{\mu\nu} = 0. \quad (6.3)$$

Since these equations are formulated in tensor form, the Maxwell equations can be easily recovered. Additionally, it is important to consider the subtle role of the **Bianchi identity**

$$\partial_\lambda F^{\mu\nu} + \partial_\mu F^{\nu\lambda} + \partial_\nu F^{\lambda\mu} = 0. \quad (6.4)$$

Gauge symmetry plays a crucial role in physics. The electromagnetic field serves as an example of a gauge field, where the Lagrangian $\mathcal{L}$ remains invariant under the gauge transformation $A_\mu \rightarrow A_\mu + \partial_\mu \lambda(x)$ (check the invariance together with equation (6.2)). This is not a global symmetry. How, then, should we interpret the redundancies introduced by gauge symmetry? The answer lies in recognizing that two states related by a gauge transformation should be identified as the same physical state. One way to see the necessity of this interpretation is to observe that Maxwell's equations alone are insufficient to fully determine the evolution of A_μ . The equations are given by

$$[\eta_{\mu\nu}(\partial^\rho \partial_\rho) - \partial_\mu \partial_\nu] A^\mu = 0, \quad (6.5)$$

but the operator $\eta_{\mu\nu}(\partial^\rho \partial_\rho)$ annihilates any functions of the form $\partial_\mu \lambda$, we cannot distinguish A_μ and $A_\mu + \partial_\mu \lambda$.

In terms of components, we have

$$-\partial_j^2 A_0 + \partial_t \partial_j A_j = 0 \quad (6.6)$$

and

$$\square A_i - \partial_i(\partial_t A_0 - \partial_j A_j) = 0. \quad (6.7)$$

If we choose Coulomb gauge $\partial_j A_j = 0$, it's apparent that $A_0 = 0$ from equation (6.6) and $\square A_i = 0$ we get $p_0 = |p|^2$ from equation (6.7). We have reduced the degrees of freedom to 2.

As what we did for scalar fields, in momentum space, we can write the electromagnetic field as

$$A(x) = \int \frac{d^3 p}{(2\pi)^3} \varepsilon(p) e^{ip \cdot x}, \quad (6.8)$$

plug in the condition $\square A_i = 0$ we get $p^2 = 0$. With the condition $\partial_i A_i = 0$, we get $\varepsilon \cdot p = 0$. We can choose a certain frame

$$p = (E, 0, 0, E), \quad \varepsilon_1 = (0, 1, 0, 0), \quad \varepsilon_2 = (0, 0, 1, 0). \quad (6.9)$$

to illustrate the theory.

To explain this more intuitively, due to Lorentz contraction along the direction of motion, the longitudinal components of the photon approach zero, causing light to become effectively horizontal.

As a result, even the mathematical structure of the photon's (little group) differs from that of massive particles.

7 Lecture 7

7.1 Quantization of Maxwell fields

Now we introduce gauge symmetry for electromagnetic field. Consider a complex scalar field $\phi = \phi_1 + i\phi_2$, we promote the global gauge transformation to a local one $\phi \rightarrow e^{i\alpha(x)}\phi$. Terms like $m\phi^*\phi$ will be invariant, but $|\partial_\mu\phi|^2$ will not and the Lagrangian will be changed. To construct an invariant Lagrangian under such transformation, we introduce gauge field A_μ that transforms along with the field

$$A_\mu \rightarrow A_\mu + \frac{1}{e}\partial_\mu\alpha(x), \quad (7.1)$$

to compensate the difference of spacetime derivative caused by $\alpha(x)$. The Lagrangian is now written as

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + (D_\mu\phi)^*(D_\mu\phi) - m^2\phi^*\phi, \quad (7.2)$$

where we have defined **covariant derivative** to be $D_\mu = \partial_\mu + ieA_\mu$. This is the Lagrangian of the theory known as **scalar QED**. It is gauge-invariant.

Following the procedure of canonical quantization, we have

$$\phi(x) = \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2\omega_p}} \sum_{j=1}^2 \left(\epsilon_j a_{p,j} e^{-ip \cdot x} + \epsilon_j^\dagger a_{p,j}^\dagger e^{ip \cdot x} \right), \quad (7.3)$$

with ϵ_j being the polarization vectors that have two components. The gauge field is also quantized:

$$A_\mu = \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2\omega_p}} \sum_{i=1}^2 \left(\vec{\epsilon}_\mu^i(p) a_{p,i} e^{-ip \cdot x} + \vec{\epsilon}_\mu^i(p) a_{p,i}^\dagger e^{ip \cdot x} \right). \quad (7.4)$$

The polarization vectors here merely take the duty of projecting out unphysical degrees of freedom, ensuring that only the two transverse polarization states are considered when describing a photon. This projection eliminates any spurious components that do not correspond to observable quantities.

Questions such as “What are the Feynman rules?” and “How do we write down the photon propagator?” naturally arise after completing the quantization procedure. We have previously derived the gauge field in the Coulomb gauge, but it lacks Lorentz invariance. Now, we turn our attention to an alternative gauge: the Lorentz gauge, given by the condition $\partial_\mu A^\mu = 0$. This condition can be imposed by modifying the Lagrangian as follows:

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} - \frac{1}{2\xi}(\partial_\mu A^\mu)^2. \quad (7.5)$$

Where ξ is a Lagrangian multiplier. The equation of motion for A^μ is

$$\square A^\nu + \left(1 - \frac{1}{\xi}\right) \partial^\nu \partial_\mu A^\mu = 0. \quad (7.6)$$

The $i\epsilon$ prescription is still available, we can easily write down momentum space photon propagator

$$\langle 0|T A_\mu(x) A_\nu(y)|0\rangle = \int \frac{d^4p}{(2\pi)^4} \frac{-i \left(g_{\mu\nu} + (\xi - 1) \frac{p_\mu p_\nu}{p^2} \right)}{p^2 + i\epsilon} e^{-ip \cdot (x-y)}. \quad (7.7)$$

Generally, we choose **Feynman gauge** $\xi = 1$, and this gauge redundancy will be proved to be unphysical in Chapter 9. These tasks can be accomplished within the framework of the path integral formalism, specifically through the Faddeev-Popov procedure, which we may discuss later.

7.2 Spinors

The concept of spin, in a strict sense, was not precisely defined. For physicists, the focus is on the representations of groups, which refer to specific embeddings of a given group element g_i into operators that act on a vector space. Often, we also refer to the vectors upon which these matrices act as representations.

To figure out the intrinsic properties of Lorentz transformations (1.13), we check infinitesimal transformations, i.e. the generators of Lorentz group, they are

$$\begin{aligned} J_x &= \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 \end{pmatrix}, J_y = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 \end{pmatrix}, J_z = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \\ K_x &= \begin{pmatrix} 0 & -1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, K_y = \begin{pmatrix} 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, K_z = \begin{pmatrix} 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 \end{pmatrix} \end{aligned} \quad (7.8)$$

By this mean, a (proper orthochronous) Lorentz transformation can be written as

$$\Lambda = \exp(i\theta_i J_i + i\beta_i K_i). \quad (7.9)$$

Furthermore, we derive the Lorentz algebra as

$$\begin{aligned} [J_i, J_j] &= i\epsilon_{ijk} J_k; \\ [J_i, K_j] &= i\epsilon_{ijk} K_k; \\ [K_i, K_j] &= -i\epsilon_{ijk} J_k. \end{aligned} \quad (7.10)$$

This defines the so-called SO(1,3) algebra. Note that the third commutator does not close. The first equation is nothing but the commutation relation of rotation generators, as we have seen in quantum mechanics. Actually, it makes up a subalgebra of Lorentz algebra. Next, with some redefinitions

$$J_i^+ = \frac{1}{2}(J_i + iK_i), \quad J_i^- = \frac{1}{2}(J_i - iK_i), \quad (7.11)$$

we represent J and K as

$$J = J_i^- + J_i^+, \quad K = i(J^- - J^+). \quad (7.12)$$

The commutation relations of J_i^+ and J_i^- are

$$\begin{aligned} [J_i^+, J_j^+] &= i\epsilon_{ijk} J_k^+; \\ [J_i^-, J_j^-] &= i\epsilon_{ijk} J_k^-; \\ [J_i^+, J_j^-] &= 0. \end{aligned} \quad (7.13)$$

This indicates us that both J^+ and J_- form a subalgebra with no mixing terms. We verified an important conclusion that SO(1,3) algebra is the direct sum of two SU(2) algebras. This decomposition reflects the fact that the Lorentz group in SO(1,3) is locally isomorphic to $SU(2) \times SU(2)$. In quantum mechanics, each irreducible representation of SU(2) is characterized by a half integer j , the dimension of the vector space is $2j + 1$. Therefore, the irreducible representation of Lorentz group is carried by 2 half integers m and n , the dimension of vector space should be $(2m + 1)(2n + 1)$.

8 Lecture 8

8.1 Dirac, Weyl and Majorana fermions

Mathematically, our result claimed in the last section is $SO(1,3) = SU(2) \oplus SU(2)$. What will come for spin-half representation $(1/2, 0)$ and $(0, 1/2)$? The objects acted by this representation are called **spinors**. In particular, they are called left-handed Weyl spinor and right-handed Weyl spinor respectively. The generators for Lorentz transformations under this representation are nothing but Pauli matrices. Under Lorentz transformations, Weyl spinors transform as

$$\begin{aligned}\psi_L &\rightarrow \exp\left(\frac{1}{2}(i\theta_j\sigma_j - \beta_j\sigma_j)\right)\psi_L; \\ \psi_R &\rightarrow \exp\left(\frac{1}{2}(i\theta_j\sigma_j + \beta_j\sigma_j)\right)\psi_R.\end{aligned}\tag{8.1}$$

We define some quantities:

$$\psi = \begin{pmatrix} \psi_L \\ \psi_R \end{pmatrix}, \quad \bar{\psi} = (\psi_L \quad \psi_R),\tag{8.2}$$

$$\gamma^\mu = \begin{pmatrix} 0 & \sigma^\mu \\ \bar{\sigma}^\mu & 0 \end{pmatrix}, \text{ with } \sigma^\mu = \begin{pmatrix} 1 & \sigma \end{pmatrix}, \quad \bar{\sigma}^\mu = \begin{pmatrix} 1 & -\sigma \end{pmatrix},\tag{8.3}$$

where γ matrices satisfy

$$\{\gamma^\mu, \gamma^\nu\} = 2g^{\mu\nu}.\tag{8.4}$$

This is called **Dirac algebra**, or sometimes **Clifford algebra**. Now we can construct Lorentz-invariant Lagrangian

$$\begin{aligned}\mathcal{L} &= i\psi_R^\dagger \sigma^\mu \partial_\mu \psi_R + i\psi_L^\dagger \bar{\sigma}^\mu \partial_\mu \psi_L - m(\psi_R^\dagger \psi_L + \psi_L^\dagger \psi_R) \\ &= \bar{\psi}(\gamma^\mu \partial_\mu - m)\psi \\ &= \bar{\psi}(i\not{\partial} - m)\psi.\end{aligned}\tag{8.5}$$

It's worth to note that the nonzero mass term m couples the left-handed and right-handed spinor fields, and this term is Lorentz-invariant. The equation of motion reads:

$$(i\gamma^\mu \partial_\mu - m)\psi = 0.\tag{8.6}$$

This is **Dirac equation**. Acting $(-i\gamma^\mu \partial_\mu - m)$ on the left and using anti-commutation relation (8.4) we recover the Klein-Gordon equation.

One can write down a specific type of γ matrices satisfying Dirac algebra, as

$$\gamma^0 = \begin{pmatrix} 0 & \mathbb{1} \\ \mathbb{1} & 0 \end{pmatrix}, \quad \gamma^i = \begin{pmatrix} 0 & \sigma^i \\ -\sigma^i & 0 \end{pmatrix},\tag{8.7}$$

where $\mathbb{1}$ is identity 2×2 matrix, and the whole matrix should be a 4×4 one.

The choice of γ matrices is not unique, merely Dirac algebra (8.4) plays the most fundamental role. The form above is **Weyl representation**. Certainly it is possible to introduce other forms, like **Majorana representation**:

$$\gamma^0 = \begin{pmatrix} 0 & \sigma^2 \\ \sigma^2 & 0 \end{pmatrix}, \quad \gamma^1 = \begin{pmatrix} i\sigma^3 & 0 \\ 0 & i\sigma^3 \end{pmatrix}, \quad \gamma^2 = \begin{pmatrix} 0 & -\sigma^2 \\ \sigma^2 & 0 \end{pmatrix}, \quad \gamma^3 = \begin{pmatrix} -i\sigma^1 & 0 \\ 0 & -i\sigma^1 \end{pmatrix}.\tag{8.8}$$

We could write down an n-dimensional representation of the Lorentz algebra. Here it is:

$$S^{\mu\nu} = \frac{i}{4} [\gamma^\mu, \gamma^\nu]. \quad (8.9)$$

Each μ and ν would give a matrix $S^{\mu\nu}$ (γ matrices are constructed by Pauli matrices), which is a generator of Lorentz transformation. We use $S^{\mu\nu}$ as Lorentz generators in spinor space. Therefore they must satisfy Lorentz algebra

$$[S^{\mu\nu}, S^{\rho\sigma}] = i(g^{\nu\rho}S^{\mu\sigma} - g^{\mu\rho}S^{\nu\sigma} - g^{\nu\sigma}S^{\mu\rho} + g^{\mu\sigma}S^{\nu\rho}). \quad (8.10)$$

It may seem complex at first, but ultimately, it is just a set of matrices on both sides. Try working through it yourself. In fact, the Lorentz algebra can be derived using Dirac algebra, regardless of the specific representation chosen. That is, equations (8.4), (8.9), and (8.10) are independent of the representation we are considering. We can immediately write down the transformation matrix

$$\Lambda_S = \exp(i\theta_{\mu\nu}S^{\mu\nu}), \quad (8.11)$$

where $\theta_{\mu\nu}$ is the transformation parameter.

It is of importance to note that the term $\bar{\psi}\psi$ is Lorentz-invariant, while $\psi^\dagger\psi$ is not. This can be easily justified, do some Lorentz transformations:

$$\psi^\dagger\gamma^0\psi \rightarrow (\psi^\dagger\Lambda_S^\dagger)\gamma^0(\Lambda_S\psi) = \psi^\dagger\gamma^0\Lambda_S^{-1}\Lambda_S\psi = \psi^\dagger\gamma^0\psi, \quad (8.12)$$

where we have used relation $\Lambda_S\gamma^0 = \gamma^0\Lambda_S^{-1}$.

Moreover, Lagrangian $\mathcal{L} = \bar{\psi}(i\gamma^\mu\partial_\mu - m)\psi$ is invariant under global transformation $\psi \rightarrow e^{-i\alpha}\psi$. We promote the global transformation to a local one, like what we did for scalar field $\alpha \rightarrow \alpha(x)$. Meanwhile, we define covariant derivative $D_\mu = \partial_\mu + ieA_\mu$ and construct a Lagrangian that is invariant under local gauge transformations:

$$\mathcal{L} = \bar{\psi}(i\not{D} - m)\psi + \frac{1}{4}F^2. \quad (8.13)$$

In a more physical sense, we have coupled spinor field (fermion) with gauge field (photon). The corresponding equation of motion is

$$(i\not{D} - m)\psi = 0. \quad (8.14)$$

Again, by acting $(i\not{D} + m)$ on the left, we recover the Klein-Gordon equation $(-\not{D}^2 - m^2)\psi = 0$. This is equivalent to

$$\left(D_\mu^2 + \frac{e}{2}F_{\mu\nu}\sigma^{\mu\nu} + m^2\right)\psi = 0. \quad (8.15)$$

Pay close attention to the second term within the parentheses. Consider the scenario where the photon is 'turned off'; in that case, we obtain the equation $\square + m^2 = 0$. This non-trivial coupling term actually reflects certain properties of the electron, and it has been experimentally verified, particularly in tests related to the electron's anomalous magnetic moment. As an aside, the conserved current $j^\mu = \bar{\psi}\gamma^\mu\psi$.

We can better understand the chiral structure of spinors. Indeed the left-handed and right-handed spinor transforms among themselves without mixing with each other:

$$\psi \rightarrow \psi + \frac{1}{2} \begin{pmatrix} (i\theta_i - \beta_i)\sigma_i & 0 \\ 0 & (i\theta_i + \beta_i)\sigma_i \end{pmatrix} \psi, \quad (8.16)$$

where $\psi = (\psi_L, \psi_R)^T$. The transformation is done by a block-diagonal matrix. The left and right

handedness is known as its **chirality**.

How to specify a certain handedness? This motivates us to construct more mathematical structures. For a given Dirac spinor, one can try to project out left/right handed spinor by

$$\gamma^5 = i\gamma^0\gamma^1\gamma^2\gamma^3. \quad (8.17)$$

Why no γ^4 ? That's not the point. In the Weyl representation,

$$\gamma^5 = \begin{pmatrix} -\mathbb{1} & 0 \\ 0 & \mathbb{1} \end{pmatrix}. \quad (8.18)$$

Then, obviously we have

$$\gamma^5 \begin{pmatrix} \psi_L \\ \psi_R \end{pmatrix} = \begin{pmatrix} -\psi_L \\ \psi_R \end{pmatrix}, \quad (8.19)$$

which means that ψ_L and ψ_R are eigenvectors with eigenvalue -1 and $+1$ of γ^5 , respectively. Furthermore, with the help of γ^5 , we define projection operators

$$\hat{P}_R = \frac{1 + \gamma^5}{2} = \begin{pmatrix} 0 & 0 \\ 0 & \mathbb{1} \end{pmatrix}, \quad \hat{P}_L = \frac{1 - \gamma^5}{2} = \begin{pmatrix} \mathbb{1} & 0 \\ 0 & 0 \end{pmatrix}. \quad (8.20)$$

This gives us

$$\hat{P}_R \begin{pmatrix} \psi_L \\ \psi_R \end{pmatrix} = \begin{pmatrix} 0 \\ \psi_R \end{pmatrix}, \quad \hat{P}_L \begin{pmatrix} \psi_L \\ \psi_R \end{pmatrix} = \begin{pmatrix} \psi_L \\ 0 \end{pmatrix}. \quad (8.21)$$

Of course projection operators are idempotent, and independent of basis, though we have shown the explicit forms solely under Weyl basis. The equation of motion is

$$\begin{pmatrix} -m & i\sigma^\mu \partial_\mu \\ i\bar{\sigma}^\mu \partial_\mu & -m \end{pmatrix} \begin{pmatrix} \psi_L \\ \psi_R \end{pmatrix} = 0. \quad (8.22)$$

In conclusion, chiral symmetry is broken at the quantum level, a phenomenon known as the **Adler-Bell-Jackiw (ABJ) anomaly** or **chiral anomaly**. This is extensively discussed in the context of the **Atiyah-Singer index theorem**[4], which serves as a profound link between the fields of analysis and topology .

Ettore Majorana : Gifted and Misterious

Ettore Majorana (1906-unknown) was an Italian theoretical physicist who worked on neutrino masses. On 25 March 1938, he disappeared under mysterious circumstances after purchasing a ticket to travel by ship from Naples to Palermo. The Majorana equation and Majorana fermions are named after him. In 2006, the Majorana Prize was established in his memory.

Ettore Majorana's disappearance remains one of the most intriguing and unresolved mysteries in the history of science. In 1938, at the age of just 32, Majorana was at the peak of his scientific career, having made groundbreaking contributions to physics, including his prediction of Majorana fermions, particles that are their own antiparticles, and his prescient work on neutrinos. Despite this, his personal life was marked by increasing isolation and declining health, culminating in his abrupt disappearance.

In March 1938, Majorana was awarded a prestigious full professorship at the University of Naples, bypassing the usual competitive examination due to his fame as a leading physicist. However, his final years were marked by erratic behavior and personal struggles. After his time in Germany, where he had worked with Werner Heisenberg and Niels Bohr, Majorana returned to Rome in poor health,

suffering from gastritis and nervous exhaustion. He withdrew from social interactions, became increasingly reclusive, and ceased publishing.

On March 26, 1938, Majorana boarded a ship to Palermo from Naples. He was last seen by his colleagues and friends in the university. What happened next remains uncertain. Some reports suggested that Majorana may have intended to travel to South America, possibly seeking to escape the mounting political tensions in Italy under Mussolini's fascist regime. Others speculated that his disappearance was related to his deteriorating mental health and desire for solitude.

In the years following his disappearance, various theories emerged. Some believed he had committed suicide, while others thought he may have entered hiding, possibly with the intention of abandoning his life as a public figure. There were even rumors suggesting that he had become involved in espionage during World War II, though there is no evidence to support this.

In the decades since, Majorana's disappearance has captured the imagination of physicists and historians alike. His case has become emblematic of the tragic figure of the brilliant, misunderstood scientist who may have withdrawn from the world due to personal or psychological reasons. Despite extensive investigations and numerous hypotheses, Majorana's fate remains a mystery to this day, adding an enigmatic and poignant chapter to the history of science.

9 Lecture 9

9.1 Quantization of Dirac fermions

It does not make sense to quantize Dirac fermions with commutation relations. In contrast, we use **anti-commutation relations**.

9.2 QED Feynman rules

This is the first time we distinctly write down the Feynman rules. Conventionally, we use solid lines for fermions and wavy lines for photons (gauge bosons). Note that the arrows on anti-particle is reversed from the particle, meanwhile an extra arrow is required to specify the momentum direction. Except for propagators, each line is equipped with a dot.

The momentum-space QED Feynman rules are as follows:

$$\text{Incoming fermion external line: } f \xrightarrow{p} \bullet = u^s(p) \quad (9.1)$$

$$\text{Incoming anti-fermion external line: } \bar{f} \xleftarrow{p} \bullet = \bar{v}^s(p) \quad (9.2)$$

$$\text{Outgoing fermion external line: } \bullet \xrightarrow{p} f = \bar{u}^s(p) \quad (9.3)$$

$$\text{Outgoing anti-fermion external line: } \bullet \xleftarrow{p} \bar{f} = v^s(p) \quad (9.4)$$

$$\text{Fermion propagator: } \bullet \xrightarrow{p} \bullet = \frac{i(\not{p} + m)}{p^2 - m^2 + i\epsilon} \quad (9.5)$$

$$\text{Incoming photon external line: } \gamma \xrightarrow{p} \bullet = \epsilon_\mu(p) \quad (9.6)$$

$$\text{Outgoing photon external line: } \xrightarrow{p} \gamma = \epsilon_\mu^*(p) \quad (9.7)$$

$$\text{Photon propagator: } \xrightarrow{p} = \frac{-i}{p^2 + i\epsilon} \left(g_{\mu\nu} - (1 - \xi) \frac{p_\mu p_\nu}{p^2} \right) \quad (9.8)$$

$$\text{QED interaction vertex: } \begin{array}{c} f \xrightarrow{\quad} \bullet \\ \quad \swarrow \text{wavy} \\ \gamma \end{array} f = ig\gamma^\mu \quad (9.9)$$

Here we denote photon as γ , do not confuse the notation with γ^μ matrices. Lorentz indices could both be written as upper or lower indices, just make sure they are contracted over when writing down the amplitudes. As discussed earlier, we often choose Feynman gauge $\xi = 1$ for photon propagator.

10 Lecture 10

10.1 Elementary processes in QED

As an example of last section, we draw the diagram of **electron self-energy**, with the same photon for initial and final states,

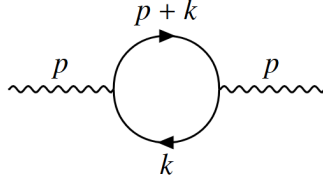


Figure 1: Feynman diagram for electron self-energy.

and write down the invariant amplitude as

$$i\mathcal{M} = \gamma^\mu \frac{i(\not{p} + \not{k} + m)}{(p+k)^2 - m^2 + i\epsilon} \gamma^\nu \frac{i(\not{k} + m)}{k^2 - m^2 + i\epsilon}. \quad (10.1)$$

But the further evaluation of this process is fairly complicated, we just forget about it for the moment.

With preparations and example, it is time to discuss some elementary and important processes in QED. Without annoying loops in diagrams, they are relatively easier to compute. The first one is the reaction $e^+e^- \rightarrow \mu^+\mu^-$. We directly draw the corresponding Feynman diagram, without writing down the QED Lagrangian or something else:

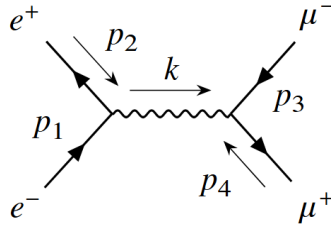


Figure 2: Feynman diagram for $e^+e^- \rightarrow \mu^+\mu^-$.

Immediately we write down the invariant amplitude according to Feynman rules:

$$i\mathcal{M} = -ie\bar{v}(p_2)\gamma^\mu u(p_1) \frac{-ig_{\mu\nu} - (1-\xi)\frac{k_\mu k_\nu}{k^2}}{k^2} (-ie)\bar{u}(p_3)\gamma^\nu v(p_4). \quad (10.2)$$

To ensure proper contraction of the matrices, the momentum of antiparticles should be written to the left of the momentum of particles. Specifically, it is crucial to write the momentum in the **opposite** direction of the arrow associated with the antiparticle. Namely, keep in mind that we must write down the momentum along the opposite arrow directions, otherwise we end up with a matrix. Remember that the momentum conservation holds at vertices:

$$k = p_1 + p_2 = p_3 + p_4. \quad (10.3)$$

For external lines, they satisfy the equations of motion

$$\begin{aligned}\not{p}_1 u(p_1) &= m u(p_1); \\ \bar{v}(p_2) \not{p}_2 &= -m \bar{v}(p_2).\end{aligned}\tag{10.4}$$

This surprisingly give us that

$$\begin{aligned}\bar{v}(p_2) \gamma^\mu u(p_1) k_\mu &= \bar{v}(p_2) \not{p}_1 u(p_1) + \bar{v}(p_2) \not{p}_2 u(p_1) \\ &= m \bar{v} u(p_1) - m \bar{v} u(p_1) \\ &= 0,\end{aligned}\tag{10.5}$$

meaning that the second term of photon propagator in equation (10.2) is erased. This indicates the fact that physical observable is independent of the choice of gauge.

We now define new quantities, the **Mandelstam variables**:

$$\begin{aligned}s &= (p + p')^2 = (k + k')^2; \\ t &= (k - p)^2 = (k' - p')^2; \\ u &= (k' - p)^2 = (k - p')^2.\end{aligned}\tag{10.6}$$

Denoting the mass of incoming and outgoing particles as m_1, m_2, m'_1 and m'_2 , it's not hard to verify that they satisfy the relation

$$s + t + u = m_1^2 + m_2^2 + m_1'^2 + m_2'^2.\tag{10.7}$$

At the same time, contract Lorentz indices, we obtain the invariant amplitude in a simpler form:

$$i\mathcal{M} = \frac{ie^2}{s} \bar{v}(p_2) \gamma^\mu u(p_1) \bar{u}(p_3) \gamma^\nu v(p_4).\tag{10.8}$$

To match experimental requirements, we square the amplitude as it is generally complex:

$$|\mathcal{M}|^2 = \frac{e^4}{m^2} [\bar{v}(p_2) \gamma^\mu u(p_1)] [\bar{u}(p_3) \gamma_\mu v(p_4)] [\bar{v}(p_4) \gamma^\nu u(p_3)] [\bar{u}(p_1) \gamma_\nu v(p_2)].\tag{10.9}$$

However, in practice, the electron (positron) beams are often unpolarized, and muon detectors are blind to the muon polarization. Hence we should average the cross section over electron and positron spin orientations, and sum the cross section over muon spin orientations. For positive muon, we have

$$\sum_s v^s(p_4) \bar{v}^s(p_4) = \not{p}_4 - m_\mu.\tag{10.10}$$

For negative muon, similarly,

$$\sum_s u^s(p_3) \bar{u}^s(p_3) = \not{p}_3 - m_\mu.\tag{10.11}$$

Sum over spins:

$$\sum_{s'} \sum_s [\bar{u}^{s'}(p_3) \gamma^\mu v^s(p_4)] [\bar{v}^s(p_3) \gamma^\mu u^{s'}(p_4)] = \text{Tr}[(\not{p}_3 + m_\mu) \gamma^\mu (\not{p}_4 - m_\mu) \gamma^\nu].\tag{10.12}$$

Do the same thing for incoming electrons, we arrive at the result that all spins are summed

$$\begin{aligned}
\sum_{\text{spins}} |\mathcal{M}|^2 &= \text{Tr}[(\not{p}_1 + m_e)\gamma^\nu(\not{p}_2 - m_e)\gamma^\mu] \text{Tr}[(\not{p}_3 + m_\mu)\gamma^\mu(\not{p}_4 - m_\mu)\gamma^\nu] \\
&= \frac{32e^4}{s^2} (p_{13}p_{24} + p_{14}p_{23} + m_\mu^2 p_{12} + m_e^2 p_{34} + 2m_e^2 m_\mu^2) \\
&= \frac{8e^4}{s^2} (t^2 + u^2 + 4s(m_e^2 + m_\mu^2) - 2(m_e^2 + m_\mu^2)^2).
\end{aligned} \tag{10.13}$$

Indeed we have omitted some convoluted trace techniques used in the first step. If possible, refer to [5] or other resources to get the full derivation procedure.

Similarly, there is another elementary process $e^-\mu^- \rightarrow e^-\mu^-$. Draw the Feynman diagram as we just did:

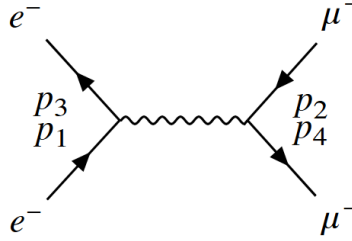


Figure 3: Feynman diagram for $e^-\mu^- \rightarrow e^-\mu^-$.

Write down the amplitude:

$$i\mathcal{M} = \frac{ie^2}{t} \bar{u}(p_3)\gamma^\mu u(p_1)\bar{u}(p_4)\gamma_\mu u(p_2), \tag{10.14}$$

where we have used Mandelstam variable t . Calculate the square amplitude and sum over spins:

$$\sum_{\text{spins}} |\mathcal{M}'|^2 = \text{Tr}[(\not{p}_1 + m_e)\gamma_\nu(\not{p}_3 + m_e)\gamma^\mu] \text{Tr}[(\not{p}_4 + m_\mu)\gamma_\mu(\not{p}_2 + m_\mu)\gamma^\nu]. \tag{10.15}$$

Upon comparing equations (10.13) and (10.15), it is evident that the two expressions are nearly identical, with the primary difference being a change in the variables used:

$$(p_1, p_2, p_3, p_4) \rightarrow (p_1, -p_3, p_4, -p_2) \quad \text{and} \quad s \rightarrow t.$$

This is an example of **crossing symmetry**, which corresponds to the exchange of Mandelstam variables. It's not a symmetry that is traditionally described by a group. You can think of the crossing symmetry this way: a particle moving along the time direction is equivalent to its antiparticle moving in the opposite direction of time. This antiparticle has negative energy and opposite momentum. Conventionally we name figure 2 and 3 as s - and t -**channel** diagrams. You can try to draw the u -channel diagram, which is topologically different from the other two. In many cases, a single scattering process may receive contributions from more than one channel.

10.2 Path integral

The **path integral** formalism is fully equivalent to the canonical quantization approach we have previously discussed. In fact, quantum field theory can be entirely formulated within the path integral framework if desired. While the path integral formalism presents certain challenges, particularly in

the computation of scattering amplitudes, it is often more advantageous than canonical quantization when it comes to evaluating correlation functions.

We should already be familiar with the path integral formulation in quantum mechanics. In the well-known double-slit experiment, the total amplitude for the two electrons is given by the sum of the contributions from each path:

$$A = A_1 + A_2. \quad (10.16)$$

Imagine a scene where infinitely many slits are distributed on infinitely many boards, the total amplitude would be calculated the same as before.

In quantum mechanics, the Hamiltonian is given by

$$\hat{H} = \frac{\hat{p}^2}{2m} + V(\hat{x}, t). \quad (10.17)$$

Start with the initial $|i\rangle = |x_i\rangle$ and final state $|f\rangle = |x_f\rangle$, if the Hamiltonian is time-independent, the amplitude matrix element would be

$$\langle f|i\rangle = \langle x_f|e^{-i\hat{H}t}|x_i\rangle. \quad (10.18)$$

Otherwise, i.e. the Hamiltonian is time-dependent, we are in a complicated situation. We make time slices $t_j = t_i + \delta t$, and insert a complete set of position eigenstates $\int dx_n |x_n\rangle \langle x_n|$ for each slice, then the amplitude is

$$\langle f|i\rangle = \int dx_n \cdots dx_1 \langle x_f|e^{-i\hat{H}(t)\delta t}|x_n\rangle \langle x_n|e^{-i\hat{H}(t)\delta t}|x_{n-1}\rangle \cdots \langle x_2|e^{-i\hat{H}(t)\delta t}|x_1\rangle \langle x_1|e^{-i\hat{H}(t)\delta t}|x_i\rangle. \quad (10.19)$$

Consider a specific slice:

$$\begin{aligned} \langle x_{j+1}|e^{-i\hat{H}(t)\delta t}|x_j\rangle &= \int \frac{dp}{2\pi} \langle x_{j+1}|p\rangle \langle p| \exp\left(-i\left(\frac{\hat{p}^2}{2m} + V(x_j, t_j)\right)\delta t\right) |x_j\rangle \\ &= \exp(-iV(x_j, t_j)\delta t) \int \frac{dp}{2\pi} \exp\left(-i\frac{\hat{p}^2}{2m}\delta t\right) \exp(i\hat{p}(x_{j+1} - x_j)) \\ &= N \exp(-iV(x_j, t_j)\delta t) \exp\left(\frac{im\delta t}{2} \frac{(x_{j+1} - x_j)^2}{(\delta t)^2}\right) \\ &= N \exp(i\mathcal{L}(x, \dot{x})). \end{aligned} \quad (10.20)$$

We have made use of $\langle p|x\rangle = e^{ipx}$ and an important Gaussian integral

$$\int dp e^{-ap^2 + Jp} = \sqrt{\frac{2\pi}{a}} e^{J^2/2a}$$

in the equation above. Each path contributes to the same amplitude, but a different phase that is proportional to the action S .

Finally, we end up with the total amplitude

$$\langle f|i\rangle = N^n \int dx_n \cdots dx_1 \exp(i\mathcal{L}(x_n, \dot{x}_n)) \cdots \exp(i\mathcal{L}(x_1, \dot{x}_1)). \quad (10.21)$$

Under the limit $\delta t \rightarrow 0$, we get

$$\langle f|i\rangle = N^n \int_{x_i}^{x_f} \mathcal{D}x(t) e^{iS[x(t)]}. \quad (10.22)$$

In quantum field theory, we are dealing with path integral from vacuum states.

$$\langle 0|0\rangle = \int \mathcal{D}\Phi_n(x) \cdots \mathcal{D}\Phi_1(x) \langle 0|e^{-i\delta t \hat{H}(t_n)}|\Phi_n\rangle \langle \Phi_n|\cdots \langle \Phi_1|e^{-i\delta t \hat{H}(t_n)}|0\rangle, \quad (10.23)$$

$$\begin{aligned} \langle \Phi_{j+1}|e^{-i\hat{H}(t_n)}|\Phi_j\rangle &= \int \mathcal{D}\Pi_j \langle \Phi_{j+1}|\Pi_j\rangle \langle \Pi_j|\exp\left(-i\delta t \int d^3x \left(\frac{\hat{\pi}^2}{2} + V(\hat{\Phi})\right)\right)|\Phi_j\rangle \\ &= N \exp\left(i\delta t \int d^3x \mathcal{L}[\phi_i, \partial_t \phi_j]\right). \end{aligned} \quad (10.24)$$

In equations above, the symbol $\mathcal{D}\Phi$ we introduced means the **Lebesgue measure** on the space of paths, it is difficult to define it in a mathematically rigorous way. Here one must use ingenuity to work with these formal expressions.

11 Lecture 11

11.1 Time-ordered correlators in path integral

Now it is time to play with correlation functions, one may curious about the meaning of

$$I = \int \mathcal{D}\Phi e^{iS} \Phi(x_j, t_j),$$

where j denotes the j th time slice. Again, we use the trick of inserting complete sets, then

$$I = \int d\Phi_n \dots d\Phi_1 \langle \Phi_f | e^{-i\hat{H}(t)\delta t} | \Phi_n \rangle \langle \Phi_n | e^{-i\hat{H}(t)\delta t} | \Phi_{n-1} \rangle \dots \langle \Phi_2 | e^{-i\hat{H}(t)\delta t} | \Phi_1 \rangle \langle \Phi_1 | e^{-i\hat{H}(t)\delta t} | \Phi_0 \rangle \Phi(x_j, t_j). \quad (11.1)$$

How to understand the R.H.S? Do not forget that Φ is a function rather than an operator in path integral. In this case, by act $\hat{\phi}(x)$ on $|\Phi\rangle$, we obtain the eigenfunction Φ :

$$\hat{\phi}(x) |\Phi\rangle = \Phi(x) |\Phi\rangle. \quad (11.2)$$

Therefore, for the j th slice, we have

$$\int \mathcal{D}\Phi_j(x) \left(e^{-iH(t_j)\delta t} |\Phi_j\rangle \langle \Phi_j| \Phi_j(x_j) \right) = \hat{\phi}(x_j) \int \mathcal{D}\Phi_j e^{-iH(t_j)\delta t} |\Phi_j\rangle \langle \Phi_j|. \quad (11.3)$$

Putting everything together,

$$\int \mathcal{D}\Phi e^{iS} \Phi(x_j, t_j) = \langle 0 | \hat{\phi}(x_j, t_j) | 0 \rangle. \quad (11.4)$$

Naturally, for two pieces of insertions,

$$\int \mathcal{D}\Phi e^{iS} \Phi(x_1, t_1) \Phi(x_2, t_2) = \langle 0 | \hat{\phi}(x_1, t_1) \hat{\phi}(x_2, t_2) | 0 \rangle. \quad (11.5)$$

If we define the relation above in the time-ordered path, following the previous derivation,

$$N \int \mathcal{D}\Phi e^{iS} \Phi(x_1, t_1) \Phi(x_2, t_2) = \langle 0 | T \hat{\phi}(x_1) \hat{\phi}(x_2) | 0 \rangle. \quad (11.6)$$

This is the time-ordered correlation function for free fields. One can define time-ordered correlation functions like this

$$\langle \Omega | T \phi(x_1) \dots \phi(x_n) | \Omega \rangle = \frac{\int \mathcal{D}\phi \phi(x_1) \dots \phi(x_n) e^{iS[\phi]}}{\int \mathcal{D}\phi e^{iS[\phi]}}. \quad (11.7)$$

We will now introduce a slicker and more formal method for computing correlation functions, based on so-called **generating functional**, avoids awkward Fourier expansions. The functional derivative $\delta/\delta J(x)$ is defined to obey the basic axiom:

$$\frac{\delta J(y)}{\delta J(x)} = \delta^4(x - y), \quad \frac{\delta}{\delta J(x)} \int d^4y J(y) \phi(y) = \phi(x).$$

For scalar field theories, the generating functional $Z[J]$ of correlation functions is defined as

$$Z[J] \equiv \int \mathcal{D}\phi \exp \left(i \int d^4x (\mathcal{L} + J(x)\phi(x)) \right). \quad (11.8)$$

We have added to $\mathcal{L}$ in the exponent a **source term**, $J(x)\phi(x)$. Correlation functions of Klein-Gordon field can be simply computed by taking functional derivatives of the generating functional. For ex-

ample, the two-point function is

$$\langle \Omega | T \phi(x_1) \phi(x_2) | \Omega \rangle = \frac{1}{Z_0} \left(-i \frac{\delta}{\delta J(x_1)} \right) \left(-i \frac{\delta}{\delta J(x_2)} \right) Z[J] \Big|_{J=0}, \quad (11.9)$$

where $Z_0 = Z[J=0]$. Each functional derivative brings down a factor of ϕ in the numerator of $Z[J]$. The calculation can be done in the same manner for n -point correlation function.

In free theory, $Z[J]$ can be computed explicitly. For example, $\mathcal{L} = -\frac{1}{2}\phi(\square + m^2)\phi$, then

$$Z[J] = \int \mathcal{D}\phi \exp \left(i \int d^4x \left(-\frac{1}{2}\phi(\square + m^2)\phi + J(x)\phi(x) \right) \right). \quad (11.10)$$

We use the following formula

$$\int_{-\infty}^{\infty} dP \exp \left(-\frac{1}{2} P A P + J P \right) = \sqrt{\frac{(2\pi)^n}{\det A}} \exp \left(\frac{1}{2} J A^{-1} J \right) \quad (11.11)$$

to complete the Gaussian integral. Be careful, we set $A = i(\square + m^2)$ above, then what's its inverse? It turns out to be the corresponding time-ordered Feynman propagator $D_F(x-y)$ (think about it!). Therefore,

$$Z[J] = Z[0] \exp \left(-\frac{1}{2} \int d^4x \int d^4y J(x) D_F(x-y) J(y) \right). \quad (11.12)$$

In this way, let's compute the two-point correlation function.

$$\begin{aligned} \langle 0 | T \phi(x_1) \phi(x_2) | 0 \rangle &= -\frac{\delta}{\delta J(x_1)} \frac{\delta}{\delta J(x_2)} \exp \left(-\frac{1}{2} \int d^4x \int d^4y J(x) D_F(x-y) J(y) \right) \Big|_{J=0} \\ &= -\frac{\delta}{\delta J(x_1)} \left(-\frac{1}{2} \int d^4y D_F(x_2-y) J(y) - \frac{1}{2} \int d^4x J(x) D_F(x-x_2) \right) \frac{Z[J]}{Z[0]} \Big|_{J=0} \\ &= D_F(x_1-x_2). \end{aligned} \quad (11.13)$$

Repeating calculation above for a few more times, the four-point function is

$$\begin{aligned} \langle 0 | T \phi_1 \phi_2 \phi_3 \phi_4 | 0 \rangle &= \frac{\delta}{\delta J_1} \frac{\delta}{\delta J_2} \frac{\delta}{\delta J_3} [J_x D_{x4}] e^{-\frac{1}{2} J_x D_{xy} J_y} \Big|_{J=0} \\ &= \frac{\delta}{\delta J_1} \frac{\delta}{\delta J_2} [-D_{34} + J_x D_{x4} J_y D_{y3}] e^{-\frac{1}{2} J_x D_{xy} J_y} \Big|_{J=0} \\ &= \frac{\delta}{\delta J_1} [D_{34} J_x D_{x2} + D_{24} J_y D_{y3} + J_x D_{x4} D_{23}] e^{-\frac{1}{2} J_x D_{xy} J_y} \Big|_{J=0} \\ &= D_{34} D_{12} + D_{24} D_{13} + D_{14} D_{23}. \end{aligned} \quad (11.14)$$

Again, this result is in agreement with what we obtained from Wick's theorem. Currently, we have managed to work out the same result under three formalisms: canonical quantization, Schwinger-Dyson equation and path integral.

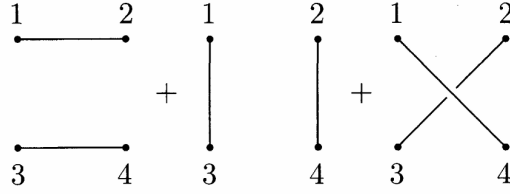


Figure 4: Three ways of contractions for four-point function.

Let's try an example of interacting theory, the Lagrangian is

$$\mathcal{L} = -\frac{1}{2}\phi(\square + m^2)\phi + \frac{1}{3!}\phi^3. \quad (11.15)$$

If coupling constant g is small enough, we perform perturbative expansion for the generating functional, carefully

$$\begin{aligned} Z[J] &= \int \mathcal{D}\phi \exp \left(-\frac{1}{2}\phi(\square + m^2)\phi + \frac{g}{3!}\phi^3 + J\phi \right) \\ &= \int \mathcal{D}\phi \exp \left(-\frac{1}{2}\phi(\square + m^2)\phi + J\phi \right) \left(1 + \frac{ig}{3!} \int d^4z \phi^3(z) + \left(\frac{ig}{3!} \right)^2 \int d^4z \int d^4w \phi^3(z)\phi^3(w) + \dots \right). \end{aligned} \quad (11.16)$$

We can take functional derivative for each term (three terms are shown above), this is the way of "order by order".

Keep the equation (11.9) in mind, we now deal with these terms one by one. The first term denotes nothing but the free theory, it gives

$$\frac{Z_0[0]}{Z[0]} D_F(x_1 - x_2).$$

The second term gives

$$\frac{Z_0[0]}{Z[0]} \frac{ig}{3!} \int d^3z \langle 0 | T \phi(x_1) \phi(x_2) \phi^3(z) | 0 \rangle.$$

So does the third term

$$\frac{Z_0[0]}{Z[0]} \left(\frac{ig}{3!} \right)^2 \int d^3z \langle 0 | T \phi(x_1) \phi(x_2) \phi^3(z) \phi^3(w) | 0 \rangle.$$

Recall the Wick's theorem that a correlator with odd number of fields ϕ gives 0, i.e. the second term vanishes. Therefore, we are confronted with a behemoth with 8 fields multiplied altogether

$$\phi_1 \phi_2 \phi_z \phi_z \phi_z \phi_z \phi_w \phi_w \phi_w.$$

Let's break it down! We get

$$\begin{aligned} \text{contractions} &= D_{12} D_{zz} D_{zw} D_{ww} \cdot 9 + D_{12} D_{zw}^3 \cdot 6 + D_{1z} D_{2z} D_{zw} D_{ww} \cdot 18 + D_{1z} D_{2w} D_{zz} D_{ww} \cdot 9 \\ &\quad + D_{1z} D_{2w} D_{zw}^2 \cdot 18 + D_{1w} D_{2z} D_{zz} D_{ww} \cdot 9 + D_{1w} D_{2z} D_{zw}^2 \cdot 18 + D_{1w} D_{2w} D_{zz} D_{zw} \cdot 18. \end{aligned} \quad (11.17)$$

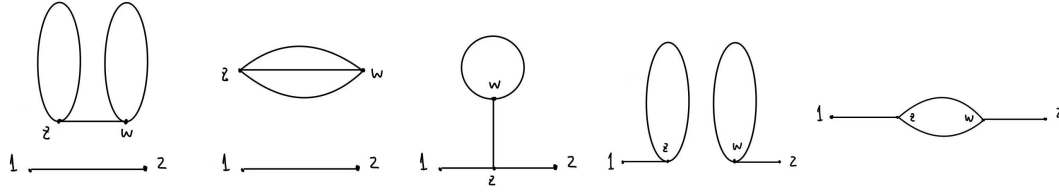


Figure 5: Ways of contractions for four-point function.

Taking account of symmetry factor, the final result is

$$\begin{aligned} \text{correlator} &= \frac{Z[0]}{Z_0[0]} \times \left(D_{12} - g^2 \int d^4z \int d^4w \right) \\ &\times \left(\frac{1}{8} \cdot D_{12} D_{zz} D_{zw} D_{ww} + \frac{1}{12} \cdot D_{12} D_{zw}^3 + \frac{1}{2} \cdot D_{1z} D_{2z} D_{zw} D_{ww} \right. \\ &\left. + \frac{1}{4} \cdot D_{1z} D_{2z} D_{ww} D_{zw} + \frac{1}{2} \cdot D_{1z} D_{zw}^2 D_{w2} \right). \end{aligned} \quad (11.18)$$

The first and second diagrams are refer to as **bubble diagram**.

As we mentioned earlier, path integral is faced with challenges for scattering problems. Generally, scattering states are constructed using the asymptotic behavior of wave functions, such as with the Lippmann-Schwinger equation. Path integrals, on the other hand, are not inherently structured to handle such scattering states directly, though they can be adapted to do so with appropriate boundary conditions or modifications.

11.2 Ward identity

Ward identity is indeed the quantum version of Noether's theorem. Following the bosonic field, we can write down the path integral for correlation function of fermionic field:

$$\langle \bar{\psi}(x_1) \psi(x_2) \rangle = \int \mathcal{D}\bar{\psi} \mathcal{D}\psi \exp \left(i \int d^4x [\bar{\psi}(x_1) (i\cancel{\partial} - m)] \psi(x_2) \right) + \bar{\psi}(x_1) \psi(x_2) + \dots, \quad (11.19)$$

where $\bar{\psi}$ and ψ are Grassmann numbers.

Under local transformations $\psi(x) \rightarrow e^{-i\alpha(x)} \psi(x)$ and $\bar{\psi}(x) \rightarrow e^{i\alpha(x)} \bar{\psi}(x)$, the Lagrangian is not invariant while the integral measure remains the same:

$$i\bar{\psi}(x) \cancel{\partial} \psi(x) \rightarrow i\bar{\psi}(x) \cancel{\partial} \psi(x) + i\bar{\psi}(x) \gamma^\mu \psi(x) \partial_\mu \alpha(x).$$

However, consider the changes of insertions in path integral,

$$\bar{\psi}(x_1) \psi(x_2) \rightarrow e^{i\alpha(x_1)} e^{-i\alpha(x_2)} \bar{\psi}(x_1) \psi(x_2),$$

when these are put altogether, the path integral (11.19) is invariant. Consequently, we obtained an equation that is invariant for α :

$$\partial_\mu \langle j^\mu \bar{\psi}(x_1) \psi(x_2) \rangle = -\delta(x - x_1) \langle \bar{\psi}(x_1) \psi(x_2) \rangle + \delta(x - x_1) \langle \bar{\psi}(x_1) \psi(x_2) \rangle, \quad (11.20)$$

where $j^\mu = \bar{\psi}(x) \gamma^\mu \psi(x)$ is the conserved current. This is **Ward identity**, and it was first used in QED by John Clive Ward and Yasushi Takahashi to relate the wave function renormalization of the electron to its vertex renormalization factor[6].

We can also derive the Schwinger-Dyson equation (5.7) in path integral formalism.

12 Lecture 12

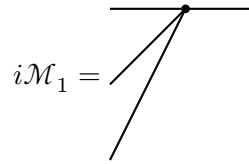
12.1 Renormalization

In Chapter 10, we have written down the amplitude for the electron self-energy (10.1) without performing the explicit calculation. This is because the expression involves a problematic loop, which leads to divergences upon evaluation. From a physical standpoint, the vacuum energy is not directly observable. To address this issue, we introduce a procedure to renormalize the theory, effectively removing the infinities and making predictions for finite, measurable quantities in terms of other observables. This procedure is known as *renormalization*. In the following, we will explore a few examples to better illustrate the underlying philosophy of this approach.

Consider a interacting theory

$$\mathcal{L} = -\frac{1}{2}\phi\Box\phi - \frac{\lambda}{4!}\phi^4,$$

at tree level, we can immediately write down the amplitude:



$$i\mathcal{M}_1 = \text{diagram} = -i\lambda. \quad (12.1)$$

For the 1-loop diagram, we have

$$i\mathcal{M}_2 = \frac{(-i\lambda)^2}{2!} \int \frac{d^4p}{(2\pi)^4} \frac{i}{k^2 + i\epsilon} \frac{i}{(p-k)^2 + i\epsilon}, \quad (12.2)$$

which we will discuss now. It is evident that this contains a logarithmic divergent integral over $1/k$. Before detailed discussion, we claim a result that

$$\partial_s \mathcal{M}(s) = -\frac{\lambda^2}{32\pi^2} \frac{1}{s}, \quad (12.3)$$

where λ is a constant, s is Mandelstam variable indicating the collision energy. Solving this differential equation, we get

$$\mathcal{M}_2 = -\frac{\lambda^2}{32\pi^2} \ln\left(\frac{s}{\Lambda^2}\right). \quad (12.4)$$

In fact, for the moment, Λ can be viewed as a momentum cutoff of the integral, i.e. the integral diverges as Λ goes to infinity. The total amplitude is the sum of the two:

$$\mathcal{M}(s) = -\lambda - \frac{\lambda^2}{32\pi^2} \ln\left(\frac{s}{\Lambda^2}\right). \quad (12.5)$$

We currently have no idea what is observable from the R.H.S.. For that, one can define a renormalized coupling constant λ_R as $-\mathcal{M}$ at a particular $s = s_0$:

$$\lambda_R \equiv -\mathcal{M}(s_0) = \lambda + \frac{\lambda^2}{32\pi^2} \ln\left(\frac{s_0}{\Lambda^2}\right). \quad (12.6)$$

We suppose that $\lambda = \lambda_R + a\lambda_R^2 + \dots$, and plug this into the equation above, we get

$$\begin{aligned} \lambda_R &= (\lambda_R + a\lambda_R^2 + \dots) + \frac{(\lambda_R + a\lambda_R^2 + \dots)^2}{32\pi^2} \ln\left(\frac{s_0}{\Lambda^2}\right) + \dots \\ &= \lambda_R + \left(a + \frac{1}{32\pi} \ln\left(\frac{s_0}{\Lambda^2}\right)\right) \lambda_R^2 + \dots \end{aligned} \quad (12.7)$$

Ignoring the quantities of higher powers, the λ_R on both sides canceled and we are end up with

$$a = \left(-\frac{1}{32\pi} \ln \left(\frac{s_0}{\Lambda^2} \right) \right). \quad (12.8)$$

Together with our definition (12.6), this gives

$$\lambda = \lambda_R - \frac{1}{32\pi^2} \lambda_R^2 \ln \left(\frac{s_0}{\Lambda} \right). \quad (12.9)$$

We are almost done! Plugging this into (12.5):

$$\begin{aligned} \mathcal{M}(s) &= - \left(\lambda_R - \frac{1}{32\pi^2} \lambda_R^2 \ln \left(\frac{s_0}{\Lambda^2} \right) \right) - \frac{\lambda_R^2}{32\pi^2} \ln \left(\frac{s}{\Lambda^2} \right) + \dots \\ &= -\lambda_R - \frac{\lambda_R^2}{32\pi^2} \ln \left(\frac{s}{s_0} \right) + \dots \end{aligned} \quad (12.10)$$

The parameter λ_R represents the coupling constant measured at an energy scale s_0 . The contributions from the momentum cutoff Λ have been effectively removed in the calculation, leading to a finite result. As a result, we have successfully achieved the objective of expressing physical observables in terms of other measurable quantities, thereby obtaining predictions for physical quantities without relying on the unphysical regularization parameters.

However, we should treat the cutoff carefully. To be precise, the momentum cutoff is a technique where we introduce an explicit upper limit on the momenta, beyond which the integrals are cut off. This means we assume that no momentum modes exist above a certain cutoff value Λ . The idea is to prevent the integral from diverging by truncating the contribution from very high-energy modes. But it is not physically meaningful in a strict sense because it introduces an artificial boundary in momentum space that does not correspond to any natural feature of the theory, though it is mathematically viable.

Now we consider the ϕ^3 theory

$$\mathcal{L} = -\frac{1}{2} \phi(\square + m^2)\phi + \frac{g}{3!} \phi^3, \quad (12.11)$$

together with the example that we have met in Chapter 5.

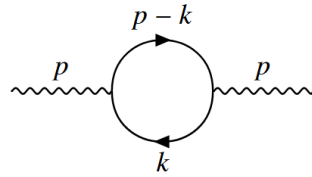


Figure 6: Feynman diagram for electron self-energy.

Following Feynman rules, we write down

$$i\mathcal{M}_{\text{loop}} = \frac{(ig)^2}{2!} \int \frac{d^4k}{(2\pi)^4} \frac{i}{(k-p)^2 - m^2 + i\epsilon} \frac{i}{k^2 - m^2 + i\epsilon}. \quad (12.12)$$

This integral is more tricky, thus some tricks are needed. We introduce the Feynman integral:

$$\frac{1}{AB} = \int_0^1 dx \frac{1}{[A + (B - A)x]^2}, \quad (12.13)$$

and let

$$A \equiv (k - p)^2 - m^2 + i\epsilon, \quad B \equiv k^2 - m^2 + i\epsilon.$$

Therefore, the original integral can be rewritten as

$$i\mathcal{M}_{\text{loop}} = \frac{g^2}{2} \int \frac{d^4k}{(2\pi)^4} \int_0^1 dx \frac{1}{[(k - p(1 - x))^2 + p^2x(1 - x) - m^2 + i\epsilon]^2}. \quad (12.14)$$

Performing variable shift $k \rightarrow k + p(1 - x)$, we further simplify the expression to something like

$$I = \int \frac{d^4k}{(2\pi)^4} \frac{1}{(k^2 - \Delta + i\epsilon)^2}. \quad (12.15)$$

We shall start with a more general case:

$$I = \int \frac{d^4k}{(2\pi)^4} \frac{1}{(k^2 - \Delta + i\epsilon)^n}.$$

The poles on the complex plane are $\pm\sqrt{k^2 + \Delta} - i\epsilon$. Choosing the proper integral contour and performing a rotation: $k_0 \rightarrow ik_0$, we have

$$k^2 \rightarrow -k_0^2 - \vec{k}^2 \equiv -k_E^2. \quad (12.16)$$

In this way, we turn a integral in Minkowski spacetime into the one in Euclidean spacetime and remove the pole since the denominator will be permanently negative. The trick to rotate the time component k_0 of the 4-momentum k by $\pi/2$ on the complex plane is called *Wick rotation*. Taking $n = 3$ case as an example, the integral can now be elegantly done:

$$i \int \frac{d^4k_E}{(2\pi)^4} \frac{1}{(-k_E^2 - \Delta + i\epsilon)^3} = \frac{-i}{32\pi^2\Delta}. \quad (12.17)$$

Take derivative of Δ in equation (12.15):

$$\partial_\Delta I(\Delta) = \int \frac{d^4k}{(2\pi)^4} \frac{2}{(k^2 - \Delta + i\epsilon)^3}. \quad (12.18)$$

Using equation (12.17) and our simplification, we eventually arrived at the final result

$$I(\Delta) = -\frac{i}{16} \ln\left(\frac{\Delta}{\Lambda^2}\right) = -\frac{i}{16} \ln\left(\frac{m^2 - p^2x(1 - x)}{\Lambda^2}\right). \quad (12.19)$$

Don't forget that it is a integral over x . However, the calculation is now simplified and more manageable:

$$\mathcal{M}_{\text{loop}} = \frac{g^2}{32\pi^2} \left(2 - \ln\left(\frac{-p^2}{\Lambda^2}\right)\right). \quad (12.20)$$

In practice, this diagram often comes as a part of another full diagram, hence beware that in general cases the external momenta is off-shell since they are then internal lines, unlike the case here. We will discuss it in a more in-depth manner.

The total amplitude should be the sum of the amplitude at tree level and one-loop correction:

$$\begin{aligned}
\mathcal{M}(Q) &= \mathcal{M}^0(Q) + \mathcal{M}^1(Q) \\
&= (ig)^2 \frac{i}{p^2} + (ig)^2 \frac{i}{p^2} i\mathcal{M}_{\text{loop}}(p) \frac{i}{p^2} \\
&= ig^2 \frac{1}{p^2} \left(-\frac{g^2}{32\pi^2} \ln \left(\frac{Q^2}{\Lambda^2} \right) \right) \frac{1}{p^2} \\
&= \frac{g^2}{Q^2} \left(1 - \frac{1}{32\pi^2} \frac{g^2}{Q^2} \ln \frac{Q^2}{\Lambda^2} \right),
\end{aligned} \tag{12.21}$$

where we have substitute variables $-p \rightarrow Q$.

To obtain a more accurate amplitude, the process of evaluating loop diagrams of higher power is necessary, though sophisticated. Physically, a loop diagram can describe the effect of virtual particle-antiparticle pairs on the propagation of a particle, leading to a correction in the particle's propagator. In some gauge theories, the computation of loop diagrams is essential for understanding quantum anomalies, such as the chiral anomaly, which can have observable consequences for particle interactions.

Next we calculate corrections of Coulomb force in QED. The most simple 1-loop diagram is

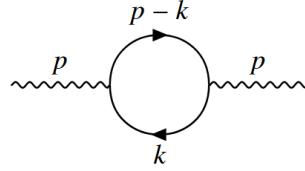


Figure 7: Feynman diagram for electron self-energy.

as we have seen in Chapter 10. Write down the invariant amplitude and sum over spins:

$$i\mathcal{M} = -(-ie^2) \int \frac{d^4k}{(2\pi)^4} \frac{i}{(p-k)^2 - m^2 + i\epsilon} \frac{i}{k^2 - m^2 + i\epsilon} \text{Tr}[\gamma^\mu (\not{k} - \not{p} + m) \gamma^\nu (\not{k} + m)]. \tag{12.22}$$

Note that the closed fermion loop contributes an extra minus sign. To get things done, we use trace technique:

$$\begin{aligned}
\text{Tr}[\gamma^\mu (\not{k} - \not{p} + m) \gamma^\nu (\not{k} + m)] &= 4(-p^\mu k^\nu - k^\mu p^\nu + 2k^\mu k^\nu + g^{\mu\nu}(-k^2 + k \cdot p + m^2)) \\
&= 4(2k^\mu k^\nu + g^{\mu\nu}(-k^2 + k \cdot p + m^2)),
\end{aligned} \tag{12.23}$$

where we have dropped the first two terms as they contribute nothing to the result (recap equation (10.2) if you are confused). Hence we are left with the integral

$$i\Pi_2^{\mu\nu} = -4e^2 \int \frac{d^4k}{(2\pi)^4} \frac{2k^\mu k^\nu + g^{\mu\nu}(-k^2 + k \cdot p + m^2)}{((p-k)^2 - m^2 + i\epsilon)(k^2 - m^2 + i\epsilon)}. \tag{12.24}$$

Taking advantage of Feynman parametrization trick and shifting $k^\mu \rightarrow k^\mu + p^\mu(1-x)$,

$$i\Pi_2^{\mu\nu} = -4e^2 \int \frac{d^4k}{(2\pi)^4} \int_0^1 dx \frac{2k^\mu k^\nu - g^{\mu\nu}(k^2 - x(1-x)p^2 - m^2)}{(k^2 + x(1-x)p^2 - m^2 + i\epsilon)^2}. \tag{12.25}$$

The divergence of this integral is even worse than before, as it is now a divergent integral over $1/k^2$.

It is necessary to introduce the method of *dimensional regularization*. The idea of dimensional regularization is very simple to state: Compute the Feynman diagram as an analytic function of the dimensionality of space time d , the final expression for any observable quantity should have a well-

defined limit as $d \rightarrow 4$. One change the integral dimension from 4 to d , which is more general. The integral becomes

$$i\Pi_2^{\mu\nu} = 4e^2\mu^{4-d}g^{\mu\nu} \int_0^1 dx \int \frac{d^d k}{(2\pi)^d} \frac{(1 - \frac{2}{d})k^2 - x(1-x)p^2 - m^2}{k^2 + p^2x(1-x) - m^2 + i\epsilon}. \quad (12.26)$$

Fortunately, the integral of d is not that impossible:

$$\int \frac{d^d k}{(2\pi)^d} \frac{k^2}{(k^2 - \Delta + i\epsilon)^2} = -\frac{2}{d} \frac{i}{(4\pi)^{2/d}} \frac{1}{\Delta^{1-d/2}} \Gamma\left(\frac{2-d}{2}\right). \quad (12.27)$$

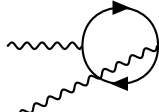
For the sake of brevity, we again let $\Delta = m^2 - px(1-x)$. Meanwhile, we have

$$\int \frac{d^d k}{(2\pi)^d} \frac{1}{(k^2 - \Delta + i\epsilon)^2} = \frac{i}{(4\pi)^{2/d}} \frac{1}{\Delta^{2-d/2}} \Gamma\left(\frac{4-d}{2}\right). \quad (12.28)$$

At last, we define $d = 4 - \epsilon$ and take the limit $\epsilon \rightarrow 0$, and eventually arrive at the final expression

$$i\Pi_2^{\mu\nu} = -\frac{e^2}{2\pi^2} p^2 g^{\mu\nu} \int_0^1 dx x(1-x) \left(\frac{2}{\epsilon} + \ln\left(\frac{\mu^2}{m^2 - p^2x(1-x)}\right) + \dots \right). \quad (12.29)$$

12.2 Running coupling



$$\begin{aligned}
 iG^{\mu\nu} &= \gamma \xrightarrow{p} + \text{diagram} \\
 &= -i \frac{G^{\mu\nu}}{p^2} + \frac{-i}{p^2} \Pi_2^{\mu\nu} \frac{-i}{p^2} \\
 &= -i \frac{(1 - e^2 \Pi_2(p^2)) g^{\mu\nu}}{p^2} + \text{terms including } p^\mu p^\nu
 \end{aligned}
 \quad (12.30)$$

If we define

$$\tilde{V}(p^2) = \frac{e^2}{p^2} (1 - e^2 \Pi_2(p^2)), \quad (12.31)$$

you may find that this is nothing but the Coulomb potential, which is something we are familiar with in general physics or classical electrodynamics.

We define the renormalized charge at p_0 such that $\tilde{p}_0 = e_R^2 p_0^{-2}$, then

$$p^2 \tilde{V}(p^2) = e^2 - e^4 \Pi_2(p^2) = e_R^4 (\Pi_2(p^2) - \Pi_2(p_0^2)). \quad (12.32)$$



Figure 8: No rush, working on it.

References

- [1] Oskar Klein. “Quantentheorie und fünfdimensionale Relativitätstheorie”. In: *Zeitschrift für Physik* 37.12 (1926), pp. 895–906.
- [2] Walter Gordon. “Der comptoneffekt nach der schrödingerschen theorie”. In: *Zeitschrift für Physik* 40.1 (1926), pp. 117–133.
- [3] Steven Weinberg. *The quantum theory of fields*. Vol. 2. Cambridge university press, 1995.
- [4] Michael Francis Atiyah and Isadore Manuel Singer. “The index of elliptic operators: I”. In: *Annals of mathematics* 87.3 (1968), pp. 484–530.
- [5] Michael E Peskin. *An introduction to quantum field theory*. CRC press, 2018.
- [6] Yasushi Takahashi. “On the generalized Ward identity”. In: *Il Nuovo Cimento* (1955-1965) 6.2 (1957), pp. 371–375.